GENERAL SCIENCE GRADE 10: LINEAR MOTION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 3.0: Natural Physical Science
Sub-Strand	Sub-Strand 3.2: Linear Motion
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Terms used in linear motion (distance, displacement, speed, velocity, acceleration) – Equations of linear motion with constant acceleration (no derivations needed) – Examples of linear motion with constant acceleration – Freely falling bodies and the effect of gravity
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain terms used in linear motion; b) calculate variables of motion using equations of linear motion with constant acceleration; c) investigate the effect of gravity on bodies under free fall; d) appreciate the significance of linear motion and free fall in real life.
Core Competencies	– Communication and Collaboration: as the learner works with peers to discuss and present in plenary the application of linear motion in real life. – Learning to Learn: as the learner carries out assignments on examples of bodies under free-fall motion and presents the findings in a group discussion.
Core Values	– Peace: the learner displays tolerance and calmness when discussing the application of linear motion in real life. – Unity: the learner displays cooperation and collaboration with others while investigating the effect of gravity on objects.
Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner formulates questions about why dropped objects speed up and how gravity influences free fall near Nairobi and other locations. – Planning and Carrying Out Investigations: as the learner drops objects from measured heights to observe increasing speed and infer the effect of gravity. – Analysing and Interpreting Data: as the learner uses collected timing data from free-fall observations to calculate acceleration due to gravity and compare with $g = 10\ m/s^2$. – Using Mathematics and Computational Thinking: as the learner solves numerical problems on linear motion using the equations of motion with constant acceleration. – Constructing Explanations: as the learner explains observed differences in the speed of objects dropped from different heights using the concept of gravitational acceleration. – Obtaining, Evaluating, and Communicating Information: as the learner researches applications of linear motion and free fall

	using digital and print media and presents findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Road Safety: increased awareness on how to minimise fatal injuries resulting from road crashes on sloping roads using the concept of free fall of objects (e.g., lorries descending the steep Kinangop escarpment or the Limuru–Nairobi road). – Life Skills Education (Self-Awareness): the learner observes safety measures when dropping objects to investigate the effect of gravity on objects.
Career Connections	– Civil and Structural Engineering: understanding linear motion and free fall is essential for designing safe roads, bridges, and buildings in Kenya's expanding infrastructure (e.g., the Nairobi Expressway, SGR). – Sports Science and Athletics: Kenyan elite runners and jumpers (e.g., athletes from the Rift Valley) rely on principles of velocity, acceleration, and displacement for performance analysis and coaching. – Aviation and Aerospace Technology: free-fall calculations and motion equations underpin pilot training and the growing Aviation track at Senior School. – Traffic and Road Safety Management: NTSA officers and road engineers apply motion equations to set speed limits and design warning systems on steep Kenyan roads. – Medicine and Biomechanics: understanding free fall and impact forces informs injury prevention and rehabilitation for falls — relevant to Kenya's hospitals and physiotherapy sector.
Focus for Lessons	This unit develops learners' ability to describe, quantify, and apply the principles of linear motion — from defining key terms to solving real-world problems using the equations of constant acceleration and investigating free fall. Instruction anchors physics concepts in Kenyan everyday contexts such as road safety on escarpment roads, the flight path of a maize cob dropped from a granary, and the sprint mechanics of Rift Valley athletes. Learners progress from qualitative understanding of motion vocabulary to quantitative problem-solving and experimental investigation of gravitational acceleration.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes? KICD KEY INQUIRY QUESTIONS: 1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Anchoring Phenomenon	A mkokoteni (hand-cart) loaded with sacks of maize is accidentally released from rest at the top of the steep Limuru–Nairobi escarpment road and rolls freely downhill. Witnesses describe how it starts moving slowly but becomes dangerously fast within seconds, eventually crashing. Students view a short video clip of a runaway vehicle on a Kenyan escarpment road and are asked: "Why does the cart keep getting faster even though no one is pushing it, and could we have predicted how fast it would be going before it crashed?" This is puzzling because the cart has no engine, yet its speed continuously increases — suggesting an invisible force (gravity) is continuously doing work on it. The phenomenon directly motivates the need to define motion terms precisely, develop equations that can predict the cart's speed and distance at any moment, and understand free fall as the purest expression of gravitational acceleration.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Motion Vocabulary: Students watch a video clip of a runaway vehicle on a steep Kenyan escarpment road. They record initial observations and predictions about why the vehicle accelerates. The teacher facilitates a whole-class discussion to surface prior knowledge and misconceptions. Students are introduced to and distinguish between distance, displacement, speed, velocity, and acceleration using the mkokoteni scenario and Kenyan athletics examples. A class Driving Question Board (DQB) is created and initial questions are posted. Lesson 2 — Equations of Linear Motion (Part 1): Using the mkokoteni context, the teacher introduces the four equations of linear

	motion with constant acceleration (presented without derivation as per KICD). Students identify the five variables (u, v, a, s, t) and practise selecting the correct equation for given problems. Worked examples use Kenyan road and athletics contexts (e.g., a matatu accelerating from a bus stop on Thika Road). Students solve structured numerical problems in pairs. Lesson 3 — Equations of Linear Motion (Part 2 — Problem Solving): Students work in small groups to solve a set of progressively challenging problems using all four equations of motion, including contexts such as a Rift Valley athlete's 400 m race, a boda-boda decelerating before a police check, and a ball thrown upward from a Mombasa rooftop. Groups present solutions on the board; the teacher uses cold-calling (minimum 3 groups per problem set) and provides 30-second wait time before taking responses. The DQB is updated with new insights. Lesson 4 — Investigating Free Fall (Predict & Observe): Students first predict: "If you drop a ripe mango and a pebble from the same height, which hits the ground first?" They record predictions individually. The class then drops objects (rubber bung, crumpled paper, stone) from a fixed height of 2 m and observes that all reach the ground at the same time (in the absence of significant air resistance). Students infer that gravity accelerates all objects equally. Data on drop height and time are recorded and used to estimate g. Lesson 5 — Freely Falling Bodies (Explain & Calculate): Students use their experimental data from Lesson 4 and the equations of linear motion (with $u = 0$, $a = g \approx 10\ m/s^2$) to calculate the speed of the mkokoteni at various points along the escarpment, the time taken for the mango to fall from the tree, and the height from which an object was dropped given its impact speed. The teacher explicitly connects each calculation back to the anchoring phenomenon. Students discuss the dangers of free fall on sloping surfaces and steep roads in Kenya. Lesson 6 — Applications, Road Safety & Real-Life Contexts: Using the equations and free-fall principles, students investigate the braking distance of a matatu travelling at different speeds on Mombasa Road. They discuss road safety (NTSA guidelines, signage on steep roads such as the Kinangop descent) and how understanding linear motion can save lives. Students research (using digital devices or print media) one additional real-life application of linear motion in Kenya (e.g., SGR train scheduling, pole-vault athletics, construction safety) and prepare a brief group presentation. Lesson 7 — Group Presentations & Model Building: Groups present their real-life application research from Lesson 6 (3 minutes per group; teacher cold-calls 2 members per group for follow-up questions). The class collaboratively builds and refines a visual explanatory model (annotated diagram or motion-time graph sequence) connecting the anchoring phenomenon — the mkokoteni on the escarpment — to all sub-strand concepts: motion terms, equations, free fall, gravity, and road safety mitigation. The DQB is reviewed and students identify which questions have been answered and which remain. Lesson 8 — Consolidation, DQB Review & Assessment: Students individually solve a structured set of problems revisiting all four equations and free fall, set in the context of the anchoring phenomenon (the mkokoteni) and supporting phenomena (Kenyan sprinter, falling mango). The class conducts a final DQB review — remaining questions are discussed and answered collectively. Students reflect in their journals on how their thinking about the anchoring phenomenon has changed since Lesson 1, articulating the role of gravity, constant acceleration, and the equations of motion in explaining and predicting the mkokoteni's dangerous descent.
Supporting Phenomena	– A ripe mango falling from a tree on a Kisumu farm — students notice it falls faster and faster rather than at a constant speed, raising the question of what determines how quickly it accelerates. – A Kenyan 100 m sprinter (e.g., a school athletics champion) filmed from the starting blocks — the video shows the athlete accelerating from rest to top speed, illustrating real-world non-uniform linear motion with measurable displacement and time

	intervals. – A water tank filling on a Nairobi rooftop where a pipe drips at constant intervals — drops photographed with a phone stroboscope show increasing spacing between successive positions, providing visual evidence of acceleration under gravity. – A matatu braking to a stop on Mombasa Road — passengers lurch forward, prompting discussion of deceleration (negative acceleration), braking distance, and road safety.

LESSON 1 (80 minutes (2 × 40-minute lessons)): Anchoring Phenomenon & Motion Vocabulary: Why Does the Mkokoteni Keep Getting Faster?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor the entire sub-strand in a compelling, locally relevant phenomenon — a runaway mkokoteni on the Limuru escarpment — and to establish precise scientific vocabulary for describing motion, so that students have the conceptual language needed to investigate, model, and ultimately predict the cart's speed and distance mathematically.
Knowledge	Explain the meaning and differences between the terms used in linear motion: distance, displacement, speed, velocity, and acceleration (KICD Sub-Strand 3.2a).
Skills	Observe and describe motion qualitatively; distinguish scalar from vector quantities; formulate initial questions about an unexplained phenomenon; collaborate to create and populate a Driving Question Board (DQB).
Attitudes	Appreciate the significance of precise scientific language in understanding and predicting real-life motion events; display Peace (tolerance and calmness) when discussing differing predictions; display Unity (cooperation) when building the class DQB.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	Lesson 1 launches the anchoring phenomenon — a mkokoteni released from rest on the steep Limuru–Nairobi escarpment road. Students watch a video clip of a runaway vehicle on a Kenyan escarpment, surface prior knowledge and misconceptions, and establish the shared vocabulary (distance, displacement, speed, velocity, acceleration) that every subsequent lesson will need. The class Driving Question Board is created, capturing the questions that the rest of the sub-strand will answer.
Safety Notes	No physical dropping of objects in this lesson. When viewing the video, remind students that the scenario involves real road danger; discuss respectfully. Ensure the classroom furniture is arranged so all students can see the projected screen without crowding or tripping hazards.

B. LESSON OVERVIEW
Students open the sub-strand by watching a short video clip of a runaway vehicle on a steep Kenyan escarpment road (used as a proxy for the mkokoteni phenomenon). They record raw observations and initial predictions about why the vehicle accelerates without an engine. The teacher facilitates a structured whole-class discussion to surface prior knowledge, surface misconceptions (e.g., "something must be pushing it"), and identify gaps. Students are then guided through a Kenya-athletics-anchored vocabulary activity to distinguish distance from displacement, speed from velocity, and to define acceleration — all in the context of the mkokoteni scenario. The lesson closes with the co-creation of the class Driving Question Board (DQB), on which students post their most pressing questions about the phenomenon. These questions will be revisited and answered across Lessons 1–6.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
1 — PREDICT	Students are seated in groups of	Display the still image and say:	Think-Pair-Share with public	Observable indicator: Every

(Anchoring Phenomenon Launch) 📹 **VIDEO:** [Intro to angle bisector theorem] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	4–5. Before the video plays, the teacher displays a still image of a steep escarpment road (Limuru–Nairobi highway) with a loaded mkokoteni at the top. Each student silently writes answers to two prompt questions in their exercise book: (1) 'If this mkokoteni is released from rest and rolls freely downhill, describe in words what you think will happen to its motion.' (2) 'Why do you think that will happen?' Students then share predictions with their table partner (Think-Pair). Three pairs are cold-called to share with the whole class. The teacher records key prediction words on the board (e.g., 'moves faster', 'gravity pulls it', 'gets heavier', 'someone must push') WITHOUT evaluating them — all ideas are honoured at this stage.	'Look carefully at this image. This is a mkokoteni — a hand-cart — loaded with sacks of maize, sitting at the top of the Limuru escarpment road. Imagine the rope holding it snaps. In your exercise book, write what you think will happen to its motion and WHY. You have 3 minutes — work silently.' [START TIMER — 3 minutes silent individual writing.] After 3 minutes: 'Now share your prediction with the person next to you. You each have 1 minute.' [WAIT — 2 minutes pair talk.] Cold-call 3 pairs using name sticks: 'Tell us what your pair predicted and one reason.' Record exact student words on the board under the heading 'OUR PREDICTIONS'. Do NOT correct misconceptions yet — say: 'I am writing all ideas — we will test them against evidence very soon.'	record. Writing before speaking forces individual commitment to a prediction, reducing groupthink. The public record on the board creates a shared reference that the class will return to at the end of the lesson to evaluate which predictions were supported by evidence — this is NGSS Science and Engineering Practice 1 (Asking Questions) and Practice 6 (Constructing Explanations).	student has written at least two sentences in their exercise book before the pair-share. Teacher circulates during silent writing and notes (mentally or on a clipboard) which students invoke 'gravity' or 'force' and which say 'it just moves' — this informs targeted questioning later. Target: ≥80% of students produce a written prediction referencing some causal agent (gravity, slope, weight, push).
2 — OBSERVE (Video Evidence Gathering) 📹 **VIDEO:** [Intro to angle bisector theorem] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher plays a 2–3 minute video clip of a runaway vehicle on a steep Kenyan escarpment road (or a simulation of a cart rolling down a ramp if the video is unavailable offline on the ARES platform). Students watch once with no recording task — just observe. The teacher then distributes the 'Phenomenon Observation Sheet' (a simple two-column table: 'What I observed' | 'What I am puzzled by'). The video plays a second time. Students write observations and puzzling things during/after the second viewing. Groups of 4–5 then share their tables and agree on their top 3 observations and top 3 puzzling things. Each group writes their top 3 puzzles on a sticky note (or a piece of paper) for later DQB posting.	Before first viewing say: 'Watch the full clip. Do not write yet — just observe everything carefully.' [PLAY VIDEO — first viewing, full clip.] 'Now I will play it again. This time, on your Observation Sheet, write in the LEFT column everything you notice about the motion — speeds, sounds, distances — and in the RIGHT column, write anything that puzzles or surprises you.' [PLAY VIDEO — second viewing.] [WAIT 2 minutes for individual writing after second viewing.] 'In your groups, share your tables. Agree on your TOP 3 observations and TOP 3 puzzles. Write the puzzles on your sticky note — one puzzle per line. You have 4 minutes.' [CIRCULATE — listen for the key puzzle: 'It has no engine / no one is pushing — why does it keep	Observe-Record-Share. The two-pass video strategy (first for holistic observation, second for structured recording) mirrors NGSS Practice 3 (Planning and Carrying Out Investigations). The 'puzzles' column is the critical move — it positions students as sense-makers who notice anomalies, not passive viewers. The sticky notes become raw material for the DQB in Phase 4.	Observable indicator: Each group's sticky note contains at least one puzzle that references the absence of an engine or the increasing speed of the cart. Teacher collects Observation Sheets at the end of the lesson to check depth of observation. Target: ≥90% of students record at least 2 distinct observations and 2 distinct puzzles.

		speeding up?'] If no group surfaces this, prompt: 'I notice no group has written about the engine yet. Does the mkokoteni have an engine?' [WAIT 10 seconds.] Cold-call 2 groups: 'Read us your top puzzle.' Affirm: 'That is exactly the question scientists ask — what FORCE is continuously acting on that cart?'		
3 — EXPLAIN (Motion Vocabulary Sensemaking) 📹 **VIDEO:** [Intro to angle bisector theorem] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher anchors new vocabulary directly in the mkokoteni scenario and in Kenyan athletics contexts (e.g., Faith Kipyegon running a 1500 m race at Nyayo Stadium). Students work through a structured guided-notes activity (provided on the ARES platform) with five terms: DISTANCE, DISPLACEMENT, SPEED, VELOCITY, ACCELERATION. For each term: (a) teacher gives a definition; (b) students apply it to a Kenyan example; (c) students apply it back to the mkokoteni on the Limuru escarpment. Example anchors used — Distance vs Displacement: 'Faith Kipyegon runs 1500 m around the Nyayo Stadium track and finishes where she started — her distance is 1500 m but her displacement is 0 m.' Students then answer: 'The mkokoteni travels 400 m straight down the escarpment road — what is its distance? What is its displacement?' Speed vs Velocity: 'A matatu travels from Nairobi to Mombasa at an average speed of 90 km/h — but velocity requires us to say 90 km/h SOUTH-EAST.' Students answer: 'The mkokoteni moves at 15 m/s down the slope — give its velocity in words.' Acceleration: 'The mkokoteni starts at rest (0 m/s) and after 5 seconds	Say: 'To answer our driving question — to PREDICT exactly how fast the mkokoteni is going before it crashes — we need a shared scientific language. I am going to introduce five terms. For each one, I will give a definition, then we will immediately use it in a Kenyan example, then apply it to our mkokoteni.' [Write the five terms on the board as column headers.] For DISTANCE: give definition, then say: 'Faith Kipyegon wins the 1500 m at Nyayo Stadium and crosses the finish line back where she started. Her distance is 1500 m. What is her displacement? Discuss with your partner — 10 seconds.' [WAIT 10 seconds.] Cold-call 2 students. Confirm: 'Displacement is zero — she ended where she started.' Then: 'Now apply this to the mkokoteni. It rolls 400 m straight downhill. What is its distance? What is its displacement? Write the answer — you have 30 seconds.' [WAIT.] Cold-call 1 student. Repeat this Teach-Apply-Transfer cycle for SPEED, VELOCITY, and ACCELERATION. For ACCELERATION specifically, say: 'Look at your prediction on the board. Several of you said the mkokoteni speeds up. The scientific name for speeding up is	Teach-Apply-Transfer cycle with Kenyan anchor examples. Each new term is immediately used in a familiar Kenyan athletics or transport context before being applied to the phenomenon, ensuring vocabulary is not abstract. This is NGSS Practice 6 (Constructing Explanations) and supports the KICD competency of Communication and Collaboration. The concept map requires students to relate terms to each other rather than memorise definitions in isolation, building relational understanding.	Observable indicator: During the concept-mapping task, teacher circulates and checks that (1) students correctly distinguish distance from displacement with a directional example, (2) students correctly distinguish speed from velocity, and (3) at least one arrow in every concept map connects 'velocity' to 'acceleration' with the phrase 'rate of change of' or equivalent. Cold-call responses during the Teach-Apply-Transfer cycles reveal misconceptions in real time. Target: ≥75% of concept maps show all five terms correctly connected with labelled arrows.

	is moving at 10 m/s — it is ACCELERATING. Something is continuously making it go faster.' Students complete a concept-mapping task connecting the five terms using arrows and linking phrases ('tells us direction', 'is the rate of change of', etc.).	ACCELERATION. Now the key question: WHO or WHAT is accelerating the mkokoteni? Write one sentence in your book.' [WAIT 15 seconds.] Cold-call 3 students using name sticks. Affirm responses that mention gravity or the slope. Then: 'This question — what causes the acceleration and can we predict it — is the ENGINE of our whole unit. Write it on a new sticky note for our DQB.' Close the vocabulary activity with the concept-mapping task: 'Use arrows to connect all five terms. Write what the arrow means on the arrow. You have 5 minutes.' Circulate and give targeted feedback.		
4 — DRIVING QUESTION BOARD (DQB) Creation 📹 **VIDEO:** Intro to angle bisector theorem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups retrieve their sticky notes from Phase 2 (top 3 puzzles) and the new sticky note generated at the end of Phase 3 (the acceleration question). The teacher has prepared a large DQB display (a section of the classroom wall or a large sheet of paper) with the driving question printed at the top: 'WHY does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?' The board is divided into columns: 'What causes the motion?', 'How do we describe the motion?', 'How do we calculate / predict the motion?', 'How can we make it safer?' Groups take turns posting their sticky notes into the appropriate column and briefly (30 seconds each) explain why they placed a question there. The class then votes (hands up) on the TWO questions they most want to answer by the end of the unit.	Reveal the DQB display and say: 'This is our Driving Question Board. It will live on this wall for the next six lessons. Everything we learn will be an answer to one of these questions.' Read the driving question aloud and ask the class to read it back chorally. Say: 'Take your sticky notes from earlier. In your group, decide which column each question belongs in. You have 2 minutes.' [WAIT 2 minutes.] Groups post notes one at a time. For each group: 'In 30 seconds, tell us why you placed THAT question in THAT column.' After all groups have posted: 'Look at the whole board. Which two questions do you MOST want to be able to answer by the end of this unit? Raise your hand for the one you choose.' Tally votes, star the top two questions. Then say: 'The KICD curriculum also asks us: How is free fall motion important in real life? And how can the dangers of free fall be mitigated? Does	Driving Question Board as a public knowledge-tracking artefact. The DQB externalises students' collective curiosity and creates accountability — questions posted by students belong to them. The column structure scaffolds the distinction between causal questions ('what causes it?'), descriptive questions ('how do we describe it?'), and predictive/mathematical questions ('how do we calculate it?') — mirroring the arc of the sub-strand. This is NGSS Practice 1 (Asking Questions) at scale. The voting step gives every student a stake in the unit's direction.	Observable indicator: Every group successfully posts at least 2 questions in appropriate columns (teacher can redirect if a question is in the wrong column, prompting: 'Is this question about describing or predicting?'). Teacher photographs the completed DQB at the end of the lesson as a baseline record. Target: The board contains at least 10 distinct questions, including at least one that explicitly mentions 'gravity', one that mentions 'speed' or 'velocity', and one that mentions 'prediction' or 'mathematics' or 'equations'.

	These are starred on the DQB. The teacher adds any KICD key inquiry questions not yet surfaced by students: 'How is free fall motion important in real life?' and 'How can the dangers of free fall in our day-to-day life be mitigated?'	anyone already have a sticky note for that?' If not, write these directly on the board with a different colour marker. Conclude: 'Every lesson, we will come back to this board and add what we have learned. By Lesson 6, we should be able to answer every question here.'		
5 — MODEL BUILDING (Initial Qualitative Motion Model) 📹 **VIDEO:** Intro to angle bisector theorem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students individually draw an INITIAL MODEL of the mkokoteni's motion in their exercise books. The model must include: (a) a simple sketch of the escarpment slope with the mkokoteni at the top; (b) arrows showing the direction of motion and what they believe is causing it; (c) a rough sketch-graph of what they think the speed of the mkokoteni looks like over time (axes labelled: x-axis = time, y-axis = speed). Students label their model 'LESSON 1 — INITIAL MODEL' and write one sentence underneath: 'My model cannot yet explain ____________ because I do not yet know ____________.' The teacher collects or photographs models. Students are told: 'You will update this model in every lesson. By Lesson 6, your model will be able to predict the exact speed and distance of the cart at any moment.'	Say: 'Science is about building models — simplified pictures that explain what we observe. Right now, you are going to draw your FIRST model of the mkokoteni's motion. You will update this model in every lesson as you learn more.' Display the three requirements on the board: '(1) Sketch the slope and the mkokoteni. (2) Draw arrows showing what is causing the motion and in what direction. (3) Sketch a rough speed-vs-time graph — does speed go up, stay the same, or go down?' [WAIT 5 minutes for individual model drawing.] Circulate and observe: do students draw a rising line on the speed-vs-time graph? Do they label an arrow 'gravity'? Give no corrections yet — say: 'This is your STARTING point. There are no wrong models today — only models that will get better.' After 5 minutes: 'Write the sentence starter at the bottom: My model cannot yet explain ___ because I do not yet know ___. Be honest — what do you NOT yet understand?' [WAIT 2 minutes.] Cold-call 2 students to read their sentence. Write their 'I do not yet know' phrases on the DQB under the appropriate column. Close the lesson: 'Those gaps you just named are exactly what this sub-strand is going to fill. Next lesson, we will start to measure motion precisely — and bring numbers	Initial Model construction with explicit incompleteness labelling. Asking students to name what their model CANNOT yet explain is a metacognitive move (NGSS Practice 2: Developing and Using Models) that prevents premature closure and keeps curiosity alive. The speed-vs-time graph sketch is a low-stakes introduction to graphical representation of motion, which will become quantitatively precise in later lessons. The 'Lesson 1 — Initial Model' label signals that this is a living document, not a finished product.	Observable indicator: Teacher collects/photographs models and checks for (1) presence of a directional arrow for gravity or 'downward force', (2) a speed-vs-time sketch that is non-horizontal (showing they expect speed to change), and (3) a completed sentence starter that identifies a genuine gap. Teacher uses this to group students for targeted support in Lesson 2. Target: ≥85% of students produce a model with all three required elements, even if the arrow labelling is imprecise.

		into our model.'		

D. TEACHER REFLECTION

After the lesson, consider: (1) Which misconceptions appeared most frequently in student predictions — did students believe an external push was needed, or did they invoke gravity unprompted? Record the 3 most common misconceptions to address explicitly in Lesson 2. (2) Were students able to distinguish displacement from distance using the Faith Kipyegon example? If not, plan a brief 5-minute re-teach at the start of Lesson 2 using a Nairobi matatu route as an alternative example. (3) Did the DQB questions cluster around 'causes' questions (what makes it move?) or 'prediction' questions (how fast will it go?)? If most questions are causal, seed more predictive questions at the start of Lesson 2. (4) Review the Initial Models: how many students drew a rising speed-vs-time line? Students who drew a flat or falling line hold a significant misconception about the nature of acceleration that must be addressed before equations are introduced. (5) Did all groups post at least 2 questions on the DQB? If any group was disengaged, consider restructuring group roles (recorder, speaker, question-generator) for Lesson 2. (6) Reflect on road safety PCI integration: did the discussion naturally connect to dangers on Kenyan escarpment roads (Limuru, Mai Mahiu, Kinangop)? If not, plan a more explicit connection in Lesson 3 when equations are introduced.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A video clip of a runaway vehicle accelerating freely down a steep Kenyan escarpment road, showing continuous increase in speed with no engine or external push; sticky-note puzzles generated by student groups about the cause of the increasing speed.
What did I learn?	The five key terms of linear motion — distance, displacement, speed, velocity, and acceleration — and their differences, anchored in the mkokoteni scenario and Kenyan athletics examples (Faith Kipyegon, Nyayo Stadium, Nairobi–Mombasa matatu). Students learned that the mkokoteni's continuous speed increase is called acceleration, and that gravity is the likely cause.
How does this explain the phenomenon?	Students cannot yet explain WHY gravity causes a constant, predictable acceleration, nor can they calculate how fast or how far the mkokoteni will travel at any given time. These gaps — captured in the DQB and in the Initial Model sentence starters — form the questions that Lessons 2–6 will answer through equations of linear motion and the concept of free fall.

LESSON 2 (40 minutes): Equations of Linear Motion (Part 1): Giving the Mkokoteni a Formula	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will be able to identify the five kinematic variables and apply the four equations of linear motion with constant acceleration to predict the speed and displacement of the mkokoteni at any point during its descent.
Knowledge	Students will state the four equations of linear motion (v = u + at; s = ut + half at squared; v squared = u squared + 2as; s = half of u plus v multiplied by t) and identify the five variables u, v, a, s, and t in each equation.
Skills	Students will select the correct equation for a given problem, substitute values correctly, and solve structured numerical problems set in Kenyan road and athletics contexts.
Attitudes	Students will appreciate that mathematics is a powerful tool for predicting dangerous real-world events such as the mkokoteni runaway, and that precision in selecting the right equation matters.
Key Inquiry Question	How is free fall motion important in real life? This lesson builds toward that question by providing the mathematical language needed to calculate exactly how fast and how far any accelerating object travels.
Purpose in Storyline	Lesson 1 established that the mkokoteni accelerates because of gravity. Lesson 2 now asks: can we predict exactly how fast it was going and how far it had travelled at any moment? The four equations of linear motion are introduced as the mathematical tools that will let students answer the driving question quantitatively in later lessons.
Safety Notes	

B. LESSON OVERVIEW

The teacher re-anchors the lesson in the mkokoteni phenomenon and poses the question: knowing the cart started from rest and accelerated, can we calculate its speed at any moment? Students first predict what information they would need to make such a calculation. The teacher then introduces the four equations of linear motion as given tools (no derivation required per KICD), carefully labelling each variable using the mkokoteni context. Worked examples progress from the mkokoteni on the Limuru escarpment to a matatu pulling away from a bus stop on Thika Road. Students practise in pairs on structured problems, with cold-calling and 30-second wait time used during the Explain phase. The lesson closes with students updating the Driving Question Board with new mathematical insights.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Classifying shapes of distributions] Source: Khan Academy (English	Students open their exercise books to a fresh page headed: What maths do we need to describe the mkokoteni's motion? The teacher displays the anchoring question on the board: The mkokoteni starts from rest and	Teacher displays the question on the board alongside a simple sketch of the Limuru escarpment with the mkokoteni at the top. Teacher says: Before I give you any new tools, I want you to think like an engineer. Write down what	Activating prior knowledge: students retrieve what they know about speed, velocity, and acceleration from Lesson 1 and attempt to connect these to the idea of prediction. The pair-share surfaces the range of prior	Teacher scans written predictions as they circulate. Key diagnostic signals: Do students recognise that time, initial speed, and acceleration are needed? Do any students already recall v = u + at from primary science?

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Uniform Distributions (PLIX) Source: CK-12 🔍 Search ARES for similar readings	rolls down the escarpment with constant acceleration. After 6 seconds, how fast is it going? Students write individual predictions in three parts: (1) What information do you think you need to solve this? (2) What operation or formula, if any, do you already know that might help? (3) What is your guess at the answer in m/s? Students have 3 minutes of silent individual thinking before sharing with a neighbour.	information you would need and whether you already have a formula that could help. You have 3 minutes of silence — no talking yet. After 3 minutes, teacher says: Share your prediction with the person next to you. You have 90 seconds. Teacher then cold-calls three pairs using name sticks and records key words from their responses on the board under the heading: What we think we need. Teacher does NOT confirm or deny predictions at this stage, saying: Hold those ideas — we are about to find out whether your instincts are right.	knowledge without exposing individuals to embarrassment.	Misconceptions to note: belief that a heavier cart goes faster, or that speed doubles every second only if someone pushes. These are flagged mentally for targeted clarification during the Explain phase.
Observe Phase 📹 **VIDEO:** Classifying shapes of distributions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Uniform Distributions (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher distributes a structured reference card (or writes on the board) showing the four equations of linear motion as established tools, presented clearly and labelled. Students observe the equations being written one at a time and annotate each variable using the mkokoteni context guided by teacher questioning. For each equation the teacher works one fully solved example live on the board: Equation 1 (v = u + at) is illustrated with the mkokoteni starting from rest (u = 0) with a = 2 m/s squared for t = 6 s, giving v = 12 m/s. Students watch and annotate in their notebooks. Equation 2 (s = ut + half at squared) is illustrated with the same mkokoteni to find how far it has rolled in 6 seconds. Equation 3 (v squared = u squared + 2as) is illustrated by finding the cart's speed after it has travelled 50 m. Equation 4 (s = half of u plus v multiplied by t) is illustrated by finding distance when u, v, and t are known. Students copy each equation, its worked example, and	Teacher writes each equation on the board slowly and says: This is not something you need to prove today — these are tools, like a panga or a jembe, that mathematicians have already sharpened for us. Our job is to learn how to use them correctly. For each equation teacher pauses and asks: Which variables does this equation NOT include? Giving 30 seconds wait time before taking responses. Teacher uses colour chalk or marker to highlight the missing variable in each equation, saying: The variable that is missing tells you when to choose this equation. Teacher connects every example to the mkokoteni: In this worked example, the mkokoteni starts from rest on the Limuru escarpment. Rest means u equals zero — write that in blue next to u. Teacher then works the matatu example: A matatu at a bus stop on Thika Road accelerates from rest at 1.5 m/s squared. What is its speed after 8 seconds and how far has it travelled? This second context shows students	Explicit variable identification: for every example, teacher writes the GIVEN list (u = ?, v = ?, a = ?, s = ?, t = ?) and crosses out the variable not required. This scaffold teaches students to read the problem before selecting an equation. Colour coding creates a visual reference students can use independently during pair work.	Teacher circulates during annotation and looks for: Are students correctly labelling u = 0 for the mkokoteni starting from rest? Are students crossing out the irrelevant variable when selecting an equation? Teacher asks targeted questions to two or three students who appear uncertain: Point to the variable we do not know and do not need — which equation is missing that variable?

	a colour-coded variable key (u = initial velocity in blue, v = final velocity in red, a = acceleration in green, s = displacement in orange, t = time in black).	the equations apply beyond the phenomenon.		
Explain Phase 📹 **VIDEO:** [Classifying shapes of distributions] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Uniform Distributions (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Students work in pairs to solve a structured problem set of five questions of increasing difficulty, all printed or dictated clearly: (1) The mkokoteni starts from rest and accelerates at 2 m/s squared down the Limuru escarpment. Find its velocity after 10 seconds. (2) How far has the mkokoteni rolled after those 10 seconds? (3) A boda-boda on Ngong Road starts from rest and reaches 15 m/s in 5 seconds. What is its acceleration? (4) A Rift Valley Athletics Club sprinter accelerates from rest at 3 m/s squared. What is her velocity after covering 54 m? (5) The mkokoteni reaches a velocity of 20 m/s after rolling 100 m from rest. What was its acceleration? For each problem, students write the GIVEN list, identify the missing variable, state which equation they have chosen and why, substitute, and solve. After pair work (8 minutes), the teacher cold-calls three pairs per problem for problems 1, 3, and 5, using name sticks. For each cold-call the teacher says: Tell us your GIVEN list first — then which equation did you choose and why? before asking for the numerical answer.	During pair work teacher circulates actively, crouching to student level, reading GIVEN lists, and asking: Which variable is missing from your problem? Do not tell them the answer. If a pair is stuck on equation selection say: Cover the answer columns and look only at which variable is missing — which equation does not have that variable? Before each cold-call teacher says: I am going to ask one person from this pair to explain their equation choice — not just the answer. Wait 30 seconds. Cold-calls specific students by name. After the student explains, teacher asks the class: Does the class agree with the equation choice? Thumbs up or thumbs down silently. Teacher then reveals the correct solution step by step on the board, narrating: We wrote the GIVEN list, we spotted the missing variable, we selected the matching equation, we substituted carefully, and we solved. Notice how the mkokoteni problem and the sprinter problem used the same equation — the physics is the same even though the contexts look different.	Structured problem-solving protocol: GIVEN list — missing variable — equation selection — substitution — solution. This metacognitive scaffold prevents random formula guessing and trains systematic thinking. Peer explanation during cold-calling deepens understanding because articulating a method consolidates it.	Teacher uses thumbs-up or thumbs-down response to gauge class-wide understanding after each cold-called solution. Teacher notes which equation (typically v squared = u squared + 2as) causes most difficulty and provides an additional short example on the board if more than one third of students show thumbs-down. Exit check: before moving to DQB phase, teacher asks all students to write on a sticky note or in their book: Write the one equation you found hardest to use today and state the variable it does not include.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [DNA as the "transforming principle"]	Students return to the class Driving Question Board created in Lesson 1. Each student receives one sticky note. They write one of the following: (a) a new question raised by today's lesson (e.g., Does the acceleration stay constant all the way down the	Teacher points to the driving question on the board: Why does the mkokoteni keep getting faster, and how can we use mathematics to predict exactly how fast and how far it travels before it crashes? Teacher says: At the start of this lesson, this question had no	The DQB update makes students metacognitive about the progress of the investigation. By categorising notes into answered, new, and still wondering, students see that science is cumulative and that each lesson adds evidence toward the central question.	Teacher reads all sticky notes quickly before selecting three to read aloud. This rapid scan identifies students who are confused about whether the equations apply only to the mkokoteni or to all constant-acceleration situations — a

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Uniform Distributions (PLIX) Source: CK-12 🔍 Search ARES for similar readings	escarpment?), (b) a connection between today's equations and the driving question (e.g., We can now calculate v at any point if we know a and t), or (c) a question about what happens when the mkokoteni reaches the bottom (e.g., What happens to v if the road flattens?). Students stick their notes in the appropriate zone of the DQB: Questions we can now answer, New questions raised, Still wondering. The teacher reads three notes aloud and facilitates a 2-minute whole-class discussion.	mathematical answer. Look at what we can do now. Teacher reads a student note that captures a correct application of the equations and says: This student has just shown that we can predict the speed of the mkokoteni at any moment — as long as we know the acceleration and time. That is new evidence toward answering our driving question. Teacher also reads one note from the Still wondering zone and says: This question will be answered in Lesson 4 when we investigate what happens when we drop objects from the same height. Keep it there.		misconception the teacher can address verbally during the reading: These equations work for any object with constant acceleration — a matatu, a sprinter, a falling mango.
Model Building Phase 📹 **VIDEO:** Classifying shapes of distributions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Uniform Distributions (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 4 minutes, students sketch a simple annotated motion diagram in their notebooks. They draw the Limuru escarpment as a slope with the mkokoteni at five positions (start, 2 s, 4 s, 6 s, 8 s). At each position they label the velocity calculated using v = u + at with a = 2 m/s squared and u = 0, producing values 0, 4, 8, 12, and 16 m/s. Students draw arrows of increasing length at each position to represent the growing velocity. Under the diagram they write one sentence: The mkokoteni gets faster because gravity gives it a constant acceleration of approximately 2 m/s squared along the slope, and the equations of linear motion allow us to predict its exact speed at any moment.	Teacher models the first two positions of the diagram on the board quickly, then says: Now complete the remaining three positions in your own books. You have 3 minutes. Teacher circulates and checks that arrow lengths are proportional and that velocity values are correct. Teacher closes the lesson by pointing at the diagram and saying: In Lesson 1 we described the mkokoteni using words — acceleration, velocity, displacement. Today we gave those words numbers. In Lesson 4 we will discover exactly what value gravity gives to a in the purest case — free fall — and then we will be able to solve the driving question completely.	Visual model building: the annotated slope diagram externalises the mathematical relationship between time and velocity, making constant acceleration visible as a pattern of evenly increasing arrow lengths. This anchors the abstract equations back to the concrete physical phenomenon and prepares students for the velocity-time graph in later lessons.	Teacher scans diagrams for two things: (1) velocity values correct at each time step, confirming correct use of v = u + at, and (2) arrows drawn with increasing length, confirming qualitative understanding of acceleration. Students who draw equal-length arrows are identified for targeted support in Lesson 3.

D. TEACHER REFLECTION

After the lesson, consider: Did students consistently write a GIVEN list before selecting an equation, or did they guess at formulas? Which of the four equations caused the most errors during pair work — if v squared = u squared + 2as caused difficulty for more than a third of the class, plan an additional worked example at the start of Lesson 3 before the group problem-solving session begins. Did the mkokoteni context remain meaningful throughout the lesson or did it feel forced during the matatu example? If students seemed disengaged from the phenomenon, open Lesson 3 by explicitly recalculating the mkokoteni speed at crash point as a motivator. Were wait times observed? If cold-calling produced immediate responses without reflection, lengthen wait time to 45 seconds in Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	The mkokoteni started from rest on the Limuru escarpment and accelerated continuously. Witnesses said it became dangerously fast within seconds. We knew from Lesson 1 that acceleration means a constant change in velocity, but we could not yet put exact numbers to it.
What did I learn?	There are four equations of linear motion that apply whenever acceleration is constant. The five variables are u (initial velocity), v (final velocity), a (acceleration), s (displacement), and t (time). We select the equation that matches the three variables we know and the one we want to find. The mkokoteni starting from rest means u = 0, which simplifies the calculations.
How does this explain the phenomenon?	Using v = u + at we calculated that a mkokoteni starting from rest with an acceleration of 2 m/s squared reaches 12 m/s after just 6 seconds — fast enough to be lethal. Using s = ut + half at squared we found it had already rolled 36 m by that point. This gives us the first mathematical evidence toward answering the driving question: we CAN predict how fast and how far the cart travels, as long as we know the acceleration. The remaining puzzle is: what is the exact value of gravitational acceleration — and we will investigate that in Lesson 4.

LESSON 3 (80 minutes (2 × 40-minute lessons)): Equations of Linear Motion — Part 2: Problem Solving

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students apply all four equations of linear motion to solve progressively challenging real-life problems — including contexts drawn from Kenyan athletics, boda-boda road safety, and free-fall — deepening their mathematical ability to predict the speed and distance of the runaway mkokoteni on the Limuru escarpment at any instant.
Knowledge	Students will know: (a) all four equations of linear motion ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \frac{1}{2}(u + v)t$); (b) the meaning and SI units of displacement (s), initial velocity (u), final velocity (v), acceleration (a), and time (t); (c) how to identify the correct equation for a given problem by listing known and unknown variables; (d) that deceleration is negative acceleration and that free-fall acceleration on Earth is $g \approx 10$ m/s².
Skills	Students will be able to: (a) correctly identify given and unknown quantities in a word problem; (b) select and apply the most appropriate equation of motion; (c) solve multi-step problems involving constant acceleration, deceleration, and free fall; (d) present and justify solutions on the board using correct scientific notation and units; (e) critique peers' solutions constructively.
Attitudes	Students will appreciate: (a) the power of mathematics to predict dangerous real-life events such as the mkokoteni crash; (b) the importance of road safety awareness on steep escarpment roads and at police check-points; (c) the value of collaborative problem-solving and respectful peer critique; (d) the relevance of physics to Kenyan athletic performance and everyday transport.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	In Lesson 1 students defined motion terms precisely; in Lesson 2 they derived and first applied the four equations. Lesson 3 consolidates and deepens that knowledge through structured group problem-solving using wholly Kenyan contexts — an athlete on the Rift Valley track, a boda-boda braking before a Nairobi police check-point, and a ball thrown upward from a Mombasa rooftop. By the end of this lesson students can calculate the exact speed and stopping distance of the mkokoteni at any chosen moment, directly answering the driving question's demand for mathematical prediction.
Safety Notes	No hazardous equipment is used in this lesson. Ensure aisles between group clusters are clear for teacher circulation. If a dropped-object demonstration is used for free-fall verification, perform it at bench height only, warn students to keep feet clear, and use a soft rubber ball rather than a hard object.

B. LESSON OVERVIEW

Groups of four students work through a structured Problem-Solving Card set containing three progressively challenging problems rooted in Kenyan contexts: (1) a Rift Valley athlete accelerating over a 400 m race, (2) a boda-boda decelerating before a police check-point on Thika Road, and (3) a ball thrown upward from the roof of a building in Mombasa. Groups record full working on A3 solution sheets and present on the board. The teacher uses cold-calling (minimum 3 groups per problem), enforces 30-second wait time before responses, and guides whole-class sensemaking after each problem. The lesson closes with students updating the Driving Question Board (DQB) with new predictive power for the mkokoteni scenario.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students view a single projected slide showing a silhouette of the mkokoteni at the top of the Limuru escarpment with three labelled data points: initial velocity u = 0 m/s, constant acceleration a = 2.5 m/s² (value from previous lesson's estimate), and elapsed time t = 8 s. Working individually and silently for 3 minutes, each student writes their prediction for (a) the cart's speed after 8 s and (b) the distance it has travelled, in their exercise books — using whichever method they currently feel most confident with. Students are NOT yet required to use a specific equation; the goal is to surface prior understanding and expose misconceptions about which equation to choose.	Display the slide and say: 'Before we solve anything together, I want to see YOUR thinking. Write a prediction for both speed and distance in the next 3 minutes. Show every step you try — even if you are not sure.' Set a visible 3-minute timer. Circulate silently; do NOT correct — note on a clipboard which equation choices appear (common errors: using v = u + at but forgetting to square for distance; confusing s = ut + ½at² with s = ½at²). After 3 minutes say: 'Please underline your prediction — we will return to it at the end of this lesson to see if your mathematics was correct.' Cold-call 2 volunteers to share predictions (do not confirm or deny): 'Amina, what speed did you predict? Baraka, what distance did you get?' Allow 10-second wait time after each question before accepting answer.	Individual Prediction Writing — by committing to a numerical answer before instruction, students activate prior knowledge of the equations and create cognitive dissonance when their answer is later confirmed or corrected, making the correct procedure more memorable.	Teacher scans predictions during circulation: observable indicators of readiness include correct identification of u, a, t in the problem; choice of v = u + at for speed and s = ut + ½at² for distance. A student who writes only a guess without any equation has not yet consolidated Lesson 2 content and will need targeted support during group work.
Phase 2 — Observe (Structured Group Problem Solving) 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX)	Each group of four receives a laminated Problem-Solving Card set. Problem 1 (Foundation): 'Wanjiru is a 400 m runner training at Eldoret Athletics Camp in the Rift Valley. She starts from rest and reaches a speed of 8 m/s with a constant acceleration of 2 m/s². (a) How long does it take her to reach 8 m/s? (b) How far does she travel during this acceleration phase?' Problem 2 (Intermediate): 'A boda-boda rider on Thika Road, Nairobi, is travelling at 15 m/s when he spots a police check-point 45 m ahead and brakes uniformly to a stop. (a) What is his deceleration? (b) How long does he take to stop? (c) Would he stop in time if the check-point were only	Distribute Problem-Solving Cards and A3 sheets. Say: 'You have 25 minutes. For every problem, start by writing GIVEN and FIND before touching an equation. This is the habit that saves you in exams — and would have helped engineers predict the mkokoteni crash.' Set a 25-minute timer visible to all. Circulate every 4–5 minutes; use targeted questions rather than answers: For a group stuck on Problem 2: 'How many unknowns do you have? Which equation has exactly those variables?' For a group that has not checked units: 'Read me your final answer. What are the units of deceleration?' For a group that finishes early: 'Can you solve Problem 3(c) using	Progressive Scaffolding with GIVEN/FIND/EQUATION Protocol — the structured written format forces students to decompose each problem before algebraic manipulation, mirroring the scientific practice of identifying variables before modelling. The three-level difficulty (Foundation → Intermediate → Extension) ensures every group experiences success while being appropriately challenged.	Observable indicators: (1) All group members are writing, not just one. (2) GIVEN and FIND sections are completed before any equation is written. (3) Units are present in every answer. (4) For Problem 2, students correctly assign a negative value to deceleration. A group that chooses v² = u² + 2as for Problem 1(a) (which requires time) has not yet mastered equation selection — the teacher should prompt: 'Does that equation contain t?'

Source: CK-12 🔍 Search ARES for similar readings	30 m away? Justify your answer.' Problem 3 (Extension): 'From the roof of a building in Mombasa (height 20 m), a security guard throws a ball vertically upward with an initial speed of 10 m/s. (a) What is the maximum height reached above the ground? (b) How long does the ball take to return to roof level? (c) What is the speed of the ball just before it hits the ground? Take $g = 10$ m/s².' Groups complete full written working on A3 sheets, listing: GIVEN / FIND / EQUATION / WORKING / ANSWER / UNIT CHECK. Groups have 25 minutes for all three problems.	energy methods as a check? What would you need to know?' Do NOT give answers directly. Note on clipboard which groups complete all three problems correctly for cold-call selection during presentations.		
Phase 3 — Explain (Board Presentations and Whole-Class Sensemaking) 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Three groups are cold-called to present one problem each at the front board, writing their full solution. The rest of the class acts as 'critical friends': they check units, identify any missing steps, and ask one clarifying question. After each presentation, the teacher facilitates a whole-class sensemaking discussion that explicitly connects the solved problem back to the mkokoteni phenomenon. For Problem 2 (boda-boda braking), the class discusses: 'If the mkokoteni had brakes that could produce the same deceleration as the boda-boda, could it have stopped before crashing?' Students calculate this extension question in their books.	After group work, say: 'We will now hear from three groups — one problem each. I will choose the group; the presenter will be chosen by me too, so everyone must understand your group's solution.' Cold-call Group A for Problem 1, Group B for Problem 2, Group C for Problem 3 (select groups that showed interesting correct or near-correct work during circulation). After each presentation: (1) Allow 30-second silent wait time for the class to check the solution before asking: 'Thumbs up if you agree with every step — thumbs sideways if you have a question.' (2) Cold-call at least 2 students with sideways thumbs: 'Chebet, what is your question for Group B?' (3) After peer questions, ask the phenomenon-linking question: After Problem 1 — 'Wanjiru reached 8 m/s over a short distance. Our mkokoteni had acceleration 2.5 m/s² — after 8 seconds, how fast is it going? Does your original prediction	Board Presentation with Critical Friends Protocol — public solution presentation requires students to articulate reasoning verbally, while the critical-friends role keeps the audience cognitively active. The explicit phenomenon-linking questions after each problem rebuild the narrative thread: every equation solved is a tool for understanding the mkokoteni crash, not an isolated exercise.	Observable indicators of understanding: (1) Presenter writes GIVEN/FIND/EQUATION/WORKING/ANSWER in sequence without prompting. (2) At least two 'critical friend' questions per problem address substance (e.g., unit error, sign of deceleration) rather than surface features. (3) When asked the phenomenon-linking question, at least 4 students can state the correct numerical answer for the mkokoteni speed after 8 s ($v = 0 + 2.5 \times 8 = 20$ m/s) and the distance ($s = ½ \times 2.5 \times 64 = 80$ m).

		match now?' After Problem 2 — 'The boda-boda needed 45 m to stop from 15 m/s. The mkokoteni is heavier and has no brakes. What does this tell us about the danger?' After Problem 3 — 'The ball was thrown UP but gravity still pulled it back. What does this tell us about the direction of gravitational acceleration in our mkokoteni scenario?' Minimum cold-call count per problem: 3 different students (presenter + 2 questioners). Total cold-calls this phase: minimum 9 students.		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each student receives one sticky note. They write ONE new thing they can now claim about the mkokoteni scenario that they could NOT claim after Lesson 2 — specifically, a quantitative statement or a refined question. Examples of target responses: 'After 8 seconds the mkokoteni is travelling at 20 m/s — fast enough to kill anyone in its path.' / 'If a = 2.5 m/s², the cart travels 80 m before reaching 20 m/s — less than the length of the escarpment.' / 'New question: what happens to acceleration if the road levels out? Does a become zero?' Students place notes in one of three DQB columns: WHAT WE CAN NOW PREDICT / WHAT WE STILL NEED TO EXPLAIN / NEW QUESTIONS. The class reads the new notes aloud and the teacher clusters related ideas.	Distribute sticky notes and say: 'The driving question asks whether we can predict exactly how fast and how far the mkokoteni travels. After today's problems, write one quantitative claim OR one refined question about the mkokoteni on your note — you must use numbers.' Allow 3 minutes of silent writing. Ask students to place notes on the DQB. Read 4–5 notes aloud: cold-call the authors to explain in one sentence. Say: 'Notice that our WHAT WE CAN NOW PREDICT column is growing — we are building the mathematical toolkit needed to fully answer the driving question. What is still in our WHAT WE STILL NEED TO EXPLAIN column?' Point to any notes about air resistance, road friction, or changing slope. Say: 'These will drive our next investigations.' Leave DQB visible for Lesson 4.	Driving Question Board — Claim-Evidence-Reasoning sticky notes — by requiring a quantitative claim linked to today's equations, the DQB makes students' growing understanding visible and creates a public record of the cumulative explanation being constructed across the eight-lesson sequence.	Observable indicator: at least 70% of sticky notes contain a number (speed, distance, or time) correctly derived from one of the four equations. A note that says only 'it goes faster' without numbers indicates the student has not yet transferred equation-solving skill to the phenomenon context.
Phase 5 — Model Building 📹 **VIDEO:** AP Physics 1 review of Forces	Students retrieve their cumulative motion model from Lesson 2 (a labelled diagram of the mkokoteni on the escarpment showing force arrows and motion symbols). They now add a MATHEMATICS	Say: 'Take out your motion model from Lesson 2. You are now going to upgrade it — add a mathematics layer that any engineer could use to predict the cart's behaviour. Use GREEN pen	Cumulative Annotated Model with Colour-Coded Certainty — the GREEN/ORANGE coding makes epistemically transparent what the class knows versus what remains unknown, reflecting the scientific	Observable indicators: (1) Model contains at least two correctly substituted equations with numerical answers and correct units. (2) GREEN annotations are consistent with problems solved

and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	LAYER: alongside the diagram they write a worked mini-calculation showing (a) the cart's speed at t = 8 s and (b) the distance covered in 8 s, using the correct equations with units. They also add a colour-coded annotation: GREEN = quantities we can calculate now; ORANGE = quantities we still need more information to calculate (e.g., the point at which it leaves the tarmac, air resistance effects). Pairs share models for a 2-minute peer review using the prompt: 'Is every number on their model traceable to a specific equation?'	for what we can calculate and ORANGE for what we still cannot.' Allow 8 minutes. Circulate: look for students who write equations without substituting values — prompt: 'Show me the number, not just the formula.' After peer review, cold-call one pair to share their model under the document camera. Ask: 'What is coloured ORANGE on your model — and what information would you need to turn it GREEN?' Accept 2–3 responses with 10-second wait time. Close by saying: 'Every green annotation on your model is a direct answer to our driving question. We now know the mkokoteni is travelling at 20 m/s after just 8 seconds — from REST. Next lesson we will investigate what happens when the road changes gradient and explore free fall more deeply.'	practice of distinguishing established evidence from open questions. The iterative model revision across lessons mirrors how scientists refine models as evidence accumulates.	today. (3) ORANGE annotations identify genuine unknowns (friction, variable slope) rather than quantities already established. A model with no ORANGE annotations suggests the student has not yet recognised the limits of constant-acceleration equations for the full mkokoteni scenario.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) EQUATION SELECTION: Did most groups correctly match the equation to the unknown without prompting? If not, consider introducing a decision flowchart ('Which equation has only ONE unknown?') at the start of Lesson 4. (2) COLD-CALL EQUITY: Review your clipboard notes — did you call on students from across gender and seating position, or were responses dominated by the same few learners? Adjust cold-call strategy for Lesson 4 to ensure quieter students are included. (3) PHENOMENON CONNECTION: When you asked the phenomenon-linking question after Problem 2 (boda-boda braking), could at least half the class calculate whether the mkokoteni could have stopped? If not, the bridge between equation practice and the driving question needs to be made more explicit — consider adding a dedicated 'Connect to the mkokoteni' sentence to every future problem card. (4) DQB QUALITY: Were sticky notes quantitative, or did many remain qualitative? If qualitative notes dominate, model a strong sticky note at the start of Lesson 4 before asking students to add new ones. (5) TIMING: The 25-minute group work window is tight for Problem 3 (vertical motion). If most groups did not reach Problem 3, consider making it a structured homework task with a partial solution scaffold provided, and open Lesson 4 with a 5-minute whole-class walkthrough.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed three worked problem contexts — a Rift Valley athlete accelerating, a boda-boda decelerating before a Nairobi police check-point, and a ball thrown upward from a Mombasa rooftop — and observed peer groups presenting full solutions on the board, including cases where deceleration was treated as negative acceleration and where vertical free-fall problems required careful sign conventions.

What did I learn?	Students learned how to systematically identify GIVEN and FIND quantities before selecting an equation; that the four equations of linear motion can predict speed, distance, and time for any constant-acceleration scenario; that deceleration is negative acceleration and free-fall uses $g = 10\ m/s^2$ downward; and that the runaway mkokoteni reaches approximately 20 m/s after only 8 seconds from rest, covering 80 m — making it lethally fast before it even reaches the base of the Limuru escarpment.
How does this explain the phenomenon?	Students explained why the mkokoteni keeps accelerating without anyone pushing it (gravity provides a continuous unbalanced force down the slope), and used the equations $v = u + at$ and $s = ut + \frac{1}{2}at^2$ to produce the first fully quantitative predictions for the cart's speed and distance — directly answering the driving question's demand to predict 'exactly how fast and how far' the cart travels before crashing.

LESSON 4 (80 minutes): Investigating Free Fall: Does Mass Matter? Dropping Objects from the Limuru Escarpment Height	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the effect of gravity on freely falling bodies by dropping objects of different masses and sizes from a fixed height of 2 m, and to use drop-height and time data to estimate the value of gravitational acceleration g.
Knowledge	Learners will be able to: (i) explain that all objects in free fall experience the same gravitational acceleration regardless of mass; (ii) state the value of $g \approx 9.8 \text{ m/s}^2$ and describe its significance; (iii) connect free fall to the continuously increasing speed of the mkokoteni on the Limuru escarpment.
Skills	Learners will be able to: (i) design and carry out a controlled drop experiment measuring height and fall time; (ii) use the equation $h = \frac{1}{2}gt^2$ to calculate g from experimental data; (iii) identify and control variables (mass, shape, air resistance) in an investigation; (iv) record, tabulate and analyse experimental data.
Attitudes	Learners will: (i) appreciate the importance of safety when investigating free-fall phenomena; (ii) demonstrate curiosity and persistence when experimental results challenge prior predictions; (iii) value the collaborative collection and honest recording of data.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	In Lessons 1–3, learners defined motion terms, derived the equations of linear motion, and practised applying them to the mkokoteni's journey down the Limuru escarpment. Lesson 4 now turns the spotlight on the invisible engine behind the cart's acceleration — gravity. By predicting whether a ripe mango or a pebble hits the ground first and then testing real objects, learners gather direct empirical evidence that gravity accelerates all masses equally. This evidence resolves a key puzzle on the Driving Question Board ('Does a heavier mkokoteni accelerate faster?') and equips learners with the experimental value of g they will substitute into the equations of linear motion in Lesson 5.
Safety Notes	1. All drops must be performed from a maximum height of 2 m only; no learner should stand on furniture. 2. Clear a 1 m radius landing zone on the floor below the drop point before each trial. 3. Use rubber bungs, crumpled paper balls and small stones only — no glass, no sharp objects. 4. Learners in the drop zone must step back before the countdown. 5. If the lesson is conducted outdoors, check for overhead obstacles and wind conditions that could deflect objects.

B. LESSON OVERVIEW

Lesson 4 is the first full evidence-gathering lesson in the free-fall sequence. It opens with an individual written prediction task ('Which hits the ground first — a ripe mango or a pebble?'), surfaces prior misconceptions via a quick class poll, and then moves immediately into a structured drop experiment. Working in groups of four, learners drop a rubber bung, a crumpled paper ball and a small stone from exactly 2 m — simultaneously and separately — and time the falls with a stopwatch. They record height and time in a class data table, calculate g using $h = \frac{1}{2}gt^2$, and compare their values with the accepted 9.8 m/s^2. The lesson closes with learners revising their Driving Question Board to record the new consensus: 'Gravity gives every falling object the same acceleration of about 9.8 m/s^2 — including our mkokoteni.' A cumulative annotated diagram of the mkokoteni mid-fall is updated to show the gravitational force arrow and the direction of acceleration.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Phase 1 — Predict 📹 **VIDEO:** Parametric equations intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Free Energy (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-sheet 'Prediction Slip' with three questions: (1) 'If you drop a ripe mango and a small pebble from the top of the Limuru escarpment wall at exactly the same time and height, which hits the ground first — or do they land together? Explain your reasoning in at least two sentences.' (2) 'Does a heavier mkokoteni roll down a slope faster than a lighter one because gravity pulls harder on heavier things? Yes / No / Not sure — explain.' (3) 'Predict: if an object is dropped from 2 m, roughly how many seconds will it take to reach the ground?' Learners work silently and individually for 5 minutes, writing their name on the slip. Slips are collected by the teacher but NOT marked — they will be returned and compared at the end of the lesson.	Before distributing slips, display the phenomenon link: 'Remember our mkokoteni rolling down the Limuru escarpment road. Witnesses said it got faster and faster all on its own — no engine, no push. We said gravity was responsible. But HOW does gravity work on different masses?' Distribute prediction slips. Say: 'Work alone. Be honest — there is no wrong answer right now. What you write will not be graded; it is your scientific starting point.' Allow 5 minutes of silent writing. Circulate and observe (do NOT correct). After 5 minutes, run a rapid show-of-hands poll: 'Hands up — who says the mango hits first?' (count aloud: 'I see 8 hands.') 'Hands up — the pebble hits first?' (count) 'Hands up — they land at the same time?' (count). Record the three counts on the board in a tally table labelled 'Our Class Predictions — Before Evidence'. Say: 'Interesting split. Let us gather evidence and see who is right — including me.'	Elicit–Confront–Resolve (ECR) opening move. By making predictions public and counting them, the teacher creates a class-level cognitive dissonance that motivates the experiment. Learners are personally invested in finding out whether their prediction is correct. The tally table on the board becomes a visible record of prior knowledge that will be explicitly revisited after the experiment.	Teacher circulates and scans prediction slips for the dominant misconception: 'heavier objects fall faster' (Aristotelian view). Tally the proportion of learners holding this view from the show-of-hands count. This determines how much time to allocate to the post-experiment discussion. Observable indicator: ≥70% of learners produce a written prediction with at least one causal explanation (e.g., 'because it is heavier' or 'because gravity is bigger on heavy things').
Phase 2 — Observe (Experiment: The Great Limuru Drop) 📹 **VIDEO:** Parametric equations intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Free Energy (Flexbook) Source: CK-12 🔍 Search ARES	Groups of four are assigned roles: Dropper (holds objects at exactly 2 m using a marked wall), Timer 1 (starts stopwatch at release), Timer 2 (stops stopwatch at impact sound), Recorder (enters data in table). Each group conducts THREE types of drop trials: Trial A — rubber bung (≈50 g) vs. crumpled paper ball (≈2 g) dropped simultaneously from the same hand; Trial B — stone (≈100 g) dropped alone, timed three times; Trial C — crumpled paper ball dropped alone, timed three times. Height is measured with a metre rule taped to the wall. Time data for Trials B and C are entered	Demonstrate the setup before groups begin: 'Watch carefully — I am holding the rubber bung at exactly 2 m — you can see the mark on the wall. I release WITHOUT throwing. The Dropper must open their hand, not flick. Timer 1, start the moment the hand opens; Timer 2, stop the moment you HEAR the impact. Ready?' Perform one demonstration drop. Then say: 'In your groups, complete Trial A first — both objects together. What do you observe?' (WAIT 10 seconds after each group's Trial A.) Cold-call 3 groups: 'Group Wangari — what did you observe in Trial A?'	Data Aggregation and Class Averaging. By pooling data from all groups into a single class table, learners see that random timing errors cancel out and the class average converges on the true value of g. This models real scientific practice (replication and averaging) and directly mirrors how Kenyan meteorologists at the Kenya Meteorological Department average readings across stations. The simultaneous drop of Trial A provides a dramatic, memorable 'Galileo moment' that anchors the conceptual claim.	Observable indicators: (1) Every group successfully records at least 3 timed trials for the stone drop. (2) Every group correctly substitutes into $g = 2h/t^2$ and obtains a value between 7.0 and 12.0 m/s^2 (accounting for human reaction-time error). (3) In Trial A, at least 80% of groups observe that the rubber bung and crumpled paper ball land at approximately the same time and record this correctly. Teacher circulates, checking that Recorders are using correct units (m for height, s for time, m/s^2 for g). Prompt groups with incorrect units: 'Check your units — what unit does

for similar readings	into a class data table on the board (one row per group). Learners also observe and note whether the rubber bung and crumpled paper ball land at the same time in Trial A. After all groups have contributed data, the class calculates g for each timing pair using $g = 2h/t^2$, averages the results, and compares to 9.8 m/s².	'Group Kipchoge — same or different?' 'Group Maathai — anything surprising?' After all trials, build the class data table on the board together. Ask: 'Now use $g = 2h/t^2$ — calculate your group's value of g. You have 4 minutes.' After 4 minutes, cold-call 4 groups to share their g values. Record all values. Ask: 'The accepted value of g is 9.8 m/s². How close are we? What might have caused differences?' (WAIT 12 seconds before taking answers.) Say: 'Notice — whether we used the heavy stone or the light paper ball, g comes out roughly the same. What does that tell us about gravity?'		acceleration have?'
Phase 3 — Explain 📹 **VIDEO:** [Parametric equations intro]() Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Free Energy (Flexbook)]() Source: CK-12 🔍 Search ARES for similar readings	Learners return to their seats for a guided explanation sequence. First, they receive a structured 'Explain It' worksheet with three scaffolded tasks: (Task 1) 'Using your class average value of g, calculate: (a) the speed of the mkokoteni after 3 seconds of free acceleration down the Limuru escarpment [use $v = gt$]; (b) the distance it travels in those 3 seconds [use $s = \frac{1}{2}gt^2$].' (Task 2) 'A farmer in Kericho drops a ripe mango from a height of 5 m to test if it bruises. How long does it take to hit the ground? Use $h = \frac{1}{2}gt^2$.' (Task 3) 'Why does the mkokoteni keep getting faster even though no one is pushing it? Write a three-sentence scientific explanation using the words: gravity, acceleration, constant force.' After 10 minutes of individual work, learners share answers in pairs ('pair-check'), then the teacher facilitates a whole-class discussion to consolidate the explanation.	Introduce the Explain phase by connecting back to the phenomenon: 'We have now measured g experimentally. Let us use it to answer our driving question.' Distribute the worksheet. Say: 'Work individually for 8 minutes — show all substitutions and units.' After 8 minutes: 'Now pair-check with the person next to you — compare Task 3 explanations especially.' Allow 3 minutes for pair discussion. Bring the class together. Cold-call 3 learners for Task 1a: 'Achieng, what speed did you calculate?' 'Kamau, same or different?' 'Fatuma, can you explain why the speed keeps increasing?' (WAIT 15 seconds after asking Fatuma.) For Task 3, cold-call 2 learners to read their three-sentence explanations aloud. After each, ask the class: 'Does this explanation use all three required words? Does it correctly say that the FORCE OF GRAVITY is constant and DOWNWARD?'	Worked Example → Student Application → Peer Verification. The teacher first demonstrates the algebraic logic ($F = mg$ cancellation) on the board, then learners apply it to familiar Kenyan contexts (mango drop in Kericho, mkokoteni calculation). Pair-checking before whole-class discussion reduces anxiety and surfaces lingering errors at scale. The three-sentence writing task (Task 3) forces learners to synthesise observation, data and theory into a coherent causal narrative — a core science practice.	Observable indicators: (1) ≥80% of learners correctly calculate $v = gt$ for Task 1a with correct units (m/s). (2) ≥75% of learners correctly rearrange $h = \frac{1}{2}gt^2$ to solve for t in Task 2 ($t = \sqrt{(2h/g)} \approx 1.01$ s). (3) In Task 3, the three-sentence explanation must include: a named force (gravity/weight), the word 'acceleration' or 'accelerates', and the phrase 'constant' or 'continuously'. Teacher marks Task 3 responses during the lesson close for misconceptions to address in Lesson 5.

		Correct any conflation of weight and mass: 'Remember — gravity pulls on every kilogram of the cart equally. A heavier cart feels more total force, but it also has more mass to accelerate. The ratio — which is acceleration — stays the same. This is why g is a constant.' Write on the board: 'F = mg → a = F/m = mg/m = g (constant)'. Say: 'This is why our mkokoteni accelerates at the same rate regardless of how many sacks of maize are loaded on it.'		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Parametric equations intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Free Energy (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class Driving Question Board is displayed (or projected). Learners individually write on sticky notes: (Green note) 'Something I now KNOW that I did not know before this lesson.' (Orange note) 'A new question this lesson has raised for me.' They post their notes in the correct columns of the DQB. The teacher reads selected notes aloud and facilitates a 5-minute class discussion to: (a) mark questions from previous lessons that today's evidence has answered; (b) identify new questions that connect to upcoming lessons (e.g., 'What if the road is not vertical — does the mkokoteni still accelerate at 9.8 m/s²?' previewing inclined plane concepts).	Direct learners to the DQB: 'Look at the questions we posted in Lessons 1, 2 and 3. Find the card that says: Does a heavier mkokoteni accelerate faster because gravity pulls harder on it? What does today's evidence tell us?' (WAIT 10 seconds.) Take 3 responses, then say: 'That question is now answered — let us move it to the ANSWERED column and write our consensus answer on it: All objects experience the same gravitational acceleration of approximately 9.8 m/s² regardless of mass.' Hand out sticky notes. Say: 'You have 3 minutes — one green note, one orange note. Be specific — do not just write I learned about gravity. Write: I learned that g = 9.8 m/s² because gravity's greater pull on heavier objects is exactly cancelled by their greater inertia.' After posting, read 4–5 selected notes aloud. For each orange question, say: 'Hold that question — write it down — we will return to it.' Specifically flag any note mentioning 'inclined surface' or 'friction' as a bridge to Lesson 5.	Driving Question Board as a Living Metacognitive Artefact. The act of physically moving a card from 'Questions' to 'Answered' gives learners a visible, satisfying record of growing understanding. Orange notes that preview future lessons create continuity across the 8-lesson sequence and help learners see science as a progressive, self-correcting process — mirroring how Kenyan engineers at KENHA (Kenya National Highways Authority) iteratively update road safety models.	Observable indicators: (1) Every learner posts at least one green and one orange note within 3 minutes. (2) Green notes reference specific content (g value, experimental observation, or equation). Vague notes ('I learned about falling') prompt the teacher to ask a follow-up question before the lesson ends. (3) The proportion of orange notes asking about inclined surfaces, friction, or air resistance signals readiness for Lesson 5 content.
Phase 5 —	Learners retrieve their cumulative	Say: 'Take out your mkokoteni	Iterative Model Revision	Observable indicators: (1) Weight

Model Building 📹 **VIDEO:** [Parametric equations intro] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Free Energy (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	annotated diagram of the mkokoteni on the Limuru escarpment road (started in Lesson 1 and updated each lesson). They add the following elements to today's revision: (1) A labelled downward arrow on the mkokoteni labelled 'Weight (W = mg) — constant force.' (2) An annotation: 'Gravitational acceleration g ≈ 9.8 m/s² — same for all masses.' (3) A small data box showing: 'After 3 s: v = 29.4 m/s, s = 44.1 m' (calculated from today's Explain worksheet). (4) A brief written note: 'Why the cart keeps accelerating: gravity exerts a constant downward force. Since no friction is assumed, there is no opposing force. Net force remains constant → constant acceleration → speed increases every second.' Learners compare their updated model with a partner and identify one element that has changed since Lesson 1.	diagram. Today we cracked open the physics of WHY it accelerates. Let us record that in our model.' Display a projected model on the board as a scaffold. Walk through the four additions: 'Add the weight arrow — point it straight DOWN, label it W = mg.' (Pause, circulate, check orientation.) 'Now add the g annotation — write the value we measured today.' 'Copy your calculated values from the worksheet into the data box — this is the PREDICTIVE POWER of our equations.' 'Finally, write the four-step causal chain in your own words.' After 6 minutes, pair-share: 'Show your partner one thing that changed in your model since Lesson 1. What does that change tell you about how your understanding has grown?' Cold-call 2 pairs: 'What changed for you since Lesson 1, Otieno?' (WAIT 12 seconds.) 'And for your partner Wanjiku?' Close by returning prediction slips: 'Read your prediction from the start of the lesson. Were you right? What changed your mind — and what was the evidence?' Allow 2 minutes of silent reflection before dismissal.	(cumulative across 8 lessons). Adding a new, quantitatively grounded layer to the same diagram each lesson makes conceptual growth visible and tangible. The four-step causal chain (constant force → constant acceleration → increasing speed → increasing distance) is the core explanatory structure of kinematics, and writing it in their own words forces learners to move from procedural equation use to genuine mechanistic understanding. Comparing the current model to Lesson 1 activates metacognitive reflection on learning progression.	arrow is drawn vertically downward (not along the slope) — a common error that the teacher corrects during circulation. (2) The g annotation states a value between 9.6 and 10.0 m/s². (3) The four-step causal chain uses all four terms: force, acceleration, speed (or velocity), distance. (4) In the pair-share, learners can articulate at least one specific conceptual change since Lesson 1 (e.g., 'I used to think heavier things fall faster — now I know g is constant for all masses'). Learners who cannot articulate a change are identified for a brief one-on-one check at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal: (1) What proportion of learners held the Aristotelian misconception ('heavier objects fall faster') at the start, and did the simultaneous drop of Trial A visibly shift their thinking — or did some learners rationalise the result away ('the heavy one landed just a tiny bit first')? (2) Were timing errors in the drop experiment so large that calculated g values were misleading? If so, consider introducing a video-timing method (slow-motion phone camera) or a longer drop height (3 m) in future iterations. (3) Did learners successfully connect today's g value to the mkokoteni scenario, or did free fall remain an abstract laboratory concept disconnected from the phenomenon? If the connection was weak, open Lesson 5 by replaying the escarpment video clip and explicitly overlaying the v = gt calculation on screen. (4) Which orange DQB notes are the most generative for Lesson 5 (equations of motion on slopes)? Select two or three to open the next lesson as 'questions we left unfinished today.' (5) Were all safety protocols observed — particularly the clearance of the landing zone before each drop? Note any near-misses and adjust the class management protocol accordingly before the next practical session.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We dropped a rubber bung, a crumpled paper ball and a small stone from exactly 2 m. In every trial where we dropped two objects simultaneously, they hit the ground at the same time — even though they had different masses. We timed individual drops and recorded the height and fall time.
What did I learn?	Using $h = \frac{1}{2}gt^2$, we calculated g from our experimental data and found a class average close to 9.8 m/s^2. This confirms that gravity gives ALL falling objects the same acceleration regardless of mass. After 3 seconds of free fall (or free rolling with no friction), the mkokoteni would be travelling at approximately 29.4 m/s — a speed that explains why witnesses described it as dangerously fast within seconds.
How does this explain the phenomenon?	Gravity exerts a downward force (weight, $W = mg$) on every object. Although a heavier mkokoteni experiences a greater gravitational force, it also has more mass to accelerate. The two effects cancel exactly ($a = F/m = mg/m = g$), so every object — light mango or heavy cart — accelerates downward at the same rate of 9.8 m/s^2. This constant acceleration is why the mkokoteni's speed increases by 9.8 m/s every second, even with no engine and no one pushing it.

LESSON 5 (80 minutes): Freely Falling Bodies — Explain & Calculate: Using g to Predict the Mkokoteni's Speed and Distance	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students use the equations of linear motion (with $u = 0$, $a = g \approx 10$ m/s²) to calculate the speed and distance travelled by freely falling and freely rolling bodies, connecting every calculation back to the mkokoteni on the Limuru escarpment.
Knowledge	Students will be able to: (i) state that for a body released from rest under gravity, $u = 0$ and $a = g \approx 10$ m/s²; (ii) identify the three equations of linear motion applicable to free fall: $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$; (iii) define freely falling bodies and explain why the mkokoteni accelerates without an engine force.
Skills	Students will be able to: (i) substitute $u = 0$, $a = 10$ m/s² and a known time or distance into the equations to calculate v or s; (ii) rearrange equations to find time of fall or height of drop given impact speed; (iii) present a worked solution with correct units, showing each step clearly.
Attitudes	Students will appreciate the real-life danger of uncontrolled free fall on Kenyan escarpment roads and motorised vehicles on steep gradients, and will demonstrate responsibility by proposing evidence-based safety measures.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	In Lesson 4 students dropped objects and measured fall times, confirming that acceleration due to gravity is approximately 10 m/s². Lesson 5 is the 'Explain & Calculate' turning point: students now wield the three equations of linear motion as predictive tools, computing exactly how fast the mkokoteni was moving at several checkpoints along the Limuru escarpment and how long it took to reach crash speed. This transforms qualitative observation into quantitative prediction — the heart of the driving question — and sets up Lesson 6 (velocity–time graphs) by giving students real numerical data to plot.
Safety Notes	No practical dropping of objects in this lesson. If a mango-drop demonstration is used, clear the area below, warn bystanders, and drop from no higher than 2 m onto a soft surface. Students must not lean out of windows. All worked examples use hypothetical or published data, not live drops from height.

B. LESSON OVERVIEW

An 80-minute Explain & Calculate lesson in which Grade 10 students apply the three equations of linear motion to freely falling bodies ($u = 0$, $a = g \approx 10$ m/s²). The lesson opens with students predicting the mkokoteni's speed at three checkpoints along the Limuru escarpment using intuition alone, then transitions to guided derivation of the free-fall forms of each equation. Through three progressively challenging worked examples — (1) speed of the mkokoteni after a known time, (2) time for a mango to fall from a tree near Kisumu, (3) height from which a stone was dropped given its impact speed — students build procedural fluency and conceptual understanding simultaneously. A Jigsaw cooperative structure gives each group ownership of one equation type. The lesson closes with students revising their Driving Question Board entries and updating their cumulative escarpment motion model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
1 — PREDICT	Students receive a printed	Before distributing cards, teacher	Anchored Prediction with Evidence	Observable indicator: Every

(15 minutes) 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	'Escarpment Checkpoint Card' showing a simplified side-profile of the Limuru–Nairobi road with three marked positions: Point A (top, where the mkokoteni is released), Point B (200 m down the slope, ~5 s after release), and Point C (500 m down, ~10 s after release). Working individually in silence (3 minutes), students write their gut-feel estimates: 'How fast (in km/h) is the mkokoteni travelling when it passes Point B? Point C?' They also write one sentence explaining their reasoning. Pairs then share estimates (2 minutes). A class show-of-hands poll records the distribution of guesses on the board in a rough table: <30 km/h / 30–60 km/h / 60–90 km/h / >90 km/h. Students note that the class cannot agree — motivating the need for a mathematical method. Students keep their Checkpoint Cards; they will return to them in Phase 3 to compare predictions with calculated values.	re-plays the 20-second clip of a runaway vehicle on a Kenyan escarpment road (same clip used in Lesson 1) and says: 'Last lesson we confirmed that gravity gives every falling object the same acceleration — roughly 10 metres per second every second. Today we are going to use that fact to calculate exactly how fast our mkokoteni was going at Point B and Point C — before we crashed it on paper.' Distribute Checkpoint Cards. Say: 'Write your estimates silently. There is no wrong answer yet — this is a prediction, not a test.' Allow 3 minutes of individual think time. WAIT: do not circulate or answer questions during this time. After pairs share, conduct the show-of-hands poll. Cold-call 3 students from different estimate bands: 'Akinyi, you said less than 30 km/h — what's your reasoning?' (10-second wait after question before accepting answer.) 'Kamau, you said over 90 km/h — walk us through your thinking.' 'Fatuma, you're in the middle band — what made you choose that range?' Record all three reasonings on the board under the heading 'Our Predictions'. Say: 'We will come back to these. Keep your cards.'	Card — Students commit to a numerical prediction tied to specific checkpoints on the phenomenon, activating prior knowledge and creating cognitive dissonance when results differ. The physical Checkpoint Card serves as a tangible anchor that students carry through the lesson, making the predict–observe–explain cycle visible and personal.	student has written at least one numerical estimate and one sentence of reasoning on their Checkpoint Card within 3 minutes. Teacher scans cards during pair-share; students who have left the card blank are prompted: 'Even a rough guess is useful — write any number that feels right to you.' The distribution of poll results is recorded on the board and revisited at the end of Phase 3 to show conceptual growth.
2 — OBSERVE / EVIDENCE GATHERING (20 minutes) 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	Students open their exercise books to the data table from Lesson 4 (fall heights and times recorded during the drop experiment). Teacher projects a summary class data table on the board showing the average values for h = 0.5 m, 1.0 m, 1.5 m, and 2.0 m. Students first verify their Lesson 4 conclusion by calculating g from the equation $h = ½gt^2$ for each row (individual work, 4 minutes). They should obtain	Open with: 'Before we do any new mathematics, let's confirm what we already know from Lesson 4.' Project the class-average data table. Say: 'Using the equation $h = ½gt^2$, rearrange to find g and calculate it for each row. You have 4 minutes — work individually.' WAIT 4 minutes. Cold-call 2 students to give their g values (one who dropped from 0.5 m, one from 2.0 m): 'Otieno, what did you get for the 0.5 m drop? Show your	Representation Mapping — Students explicitly connect the abstract symbols in each equation to the physical labels on the mkokoteni diagram and the Kisumu mango scenario. By annotating u = 0 and watching the equations simplify, students experience algebra as a descriptive language rather than a set of arbitrary rules, building the conceptual bridge between the phenomenon and the	Observable indicator: Every student's book contains the three simplified free-fall equations ($v = at$; $s = ½at^2$; $v^2 = 2as$) with at least one symbol annotated. Teacher circulates during the 4-minute g-calculation and checks that students are substituting correctly into $h = ½gt^2 \rightarrow g = 2h/t^2$. Students who substitute incorrectly are redirected with: 'What is h in that row? What is t? Now put them into the formula — which quantity are

📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	values clustered around 9.8–10.2 m/s². Teacher then displays a short, annotated diagram (projected or drawn on board) of the mkokoteni at the top of the Limuru escarpment, labelling: initial velocity u = 0 m/s (released from rest), acceleration a = g = 10 m/s² (down the slope, gravity component), distance s, time t, final velocity v. Students copy this diagram into their books and label it. Teacher then writes the three equations of linear motion on the board — $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$ — and students annotate each with: 'What does each symbol mean for the mkokoteni?' (u = 0 for our problem, so equations simplify to $v = at$; $s = \frac{1}{2}at^2$; $v^2 = 2as$). Students write the simplified free-fall forms beside each equation (3 minutes). Finally, teacher displays a real photograph of a mango tree on the shores of Lake Victoria near Kisumu and poses the second scenario: 'A ripe mango falls from a branch 5 m above the ground. How long does it take to hit the ground, and how fast is it moving when it lands?' Students copy the scenario into their books.	working on the board.' After both values are on the board, say: 'Our values are close to 10 m/s² — small errors come from reaction time in timing. For all our calculations today we will use g = 10 m/s² exactly.' Draw the mkokoteni diagram slowly, narrating each label: 'The cart starts from rest — that means u equals zero. Gravity pulls it down the slope — we model this as a constant acceleration of 10 metres per second squared.' Say: 'When u = 0, look what happens to our three equations.' Pause 10 seconds. Cold-call: 'Njeri, if u is zero, what does $v = u + at$ become?' Confirm $v = at$. Repeat for the other two equations. Then introduce the Kisumu mango scenario with the photograph: 'Here is a second free-fall problem — closer to home. A mango falls from 5 metres up. How long? How fast at impact?' Say: 'Copy both scenarios into your books. We will solve them step by step together in the next phase.'	mathematics.	you solving for?'
3 — EXPLAIN / SENSEMAKING (25 minutes) 📹 **VIDEO:** [Video: Plinko Probability] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations:	The class uses a Jigsaw structure. Students are in home groups of three. Each member of the home group is assigned one equation: Member A owns $v = at$ (velocity–time equation); Member B owns $s = \frac{1}{2}at^2$ (displacement–time equation); Member C owns $v^2 = 2as$ (velocity–displacement equation). Expert groups (all As together, all Bs together, all Cs together) spend 7 minutes working through their assigned equation applied to BOTH the mkokoteni	Assign roles clearly: 'Every group of three — Person sitting nearest the window is Expert A. Middle person is Expert B. Person nearest the door is Expert C.' Post three large equation cards on the board with the equation and a worked example skeleton. Say: 'In your expert groups, you have 7 minutes. Solve both problems for your equation and write your teaching card. If you finish early, write a sentence connecting your answer back to the mkokoteni.'	Jigsaw Cooperative Learning with Teaching Cards — Each student develops deep ownership of one equation by teaching it, while the home-group reassembly ensures all three equations are integrated. The Checkpoint Card then forces students to select the appropriate equation for a given scenario — a higher-order application skill. The Prediction vs. Calculation reflection sentence makes conceptual growth explicit and personal.	Observable indicators: (1) Expert group teaching cards contain a correct worked solution with units for both the mkokoteni and mango problems. (2) Checkpoint Cards show correct calculated values (50 m/s / 180 km/h at Point B; ~100 m/s / 360 km/h at Point C) with working shown. (3) Reflection sentences articulate the direction and magnitude of the prediction error. Teacher collects 5 randomly chosen Checkpoint Cards at the end of this phase for quick written

Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	scenario ($a = 10$ m/s²) and the Kisumu mango scenario. They must solve at least two numerical problems using their equation and write a 'teaching card' — a half-page worked solution they will explain back to their home group. Specific problems: Expert A: (i) Find v of the mkokoteni after t = 5 s (Answer: 50 m/s = 180 km/h); (ii) Find v of the mango just before it hits the ground, given it falls for t = 1 s (Answer: 10 m/s). Expert B: (i) Find s travelled by the mkokoteni in t = 5 s (Answer: 125 m); (ii) Find t for the mango to fall h = 5 m (Answer: 1 s). Expert C: (i) Find v of the mkokoteni after s = 200 m (Answer: 63.2 m/s ≈ 63 m/s); (ii) Find height from which a stone was dropped if its impact speed was v = 20 m/s (Answer: h = 20 m). Students return to home groups (8 minutes) and each expert teaches their equation and solutions. Home groups then complete the Checkpoint Card together: using the correct equations, calculate the mkokoteni's speed at Point B (after 5 s: v = 50 m/s = 180 km/h) and Point C (after s = 500 m: v = 100 m/s = 360 km/h). Students compare calculated values with their original predictions and write a 'Prediction vs. Calculation' reflection sentence: 'I predicted ___ km/h. The equation gives ___ km/h. This means ___.'	WAIT — circulate to expert groups, not home groups. Check Expert A groups first (they have the conceptually simplest equation and will finish earliest). Prompt struggling experts: 'What is u in this problem? Good — so what does v = u + at become? Now substitute a = 10 and your t value.' At 7 minutes, call time: 'Return to home groups. Experts, you have 8 minutes to teach your equation. Listeners, your job is to copy the worked solution and ask one question.' After Jigsaw, call the whole class together. Cold-call one student per equation to give the final answer: 'Mwangi, using Expert B's equation, how far had the mkokoteni travelled in 5 seconds?' (10-second wait.) Confirm each answer. Then say: 'Now complete your Checkpoint Card calculations with your home group.' Allow 5 minutes. Cold-call: 'Zawadi, what speed did you get at Point B? Convert it to km/h — is that speed dangerous on a Kenyan road?' Then: 'Amina, what about Point C — 500 metres down the hill? What does that tell us about why the mkokoteni crashed?' Close Phase 3 with the reflection sentence: 'Everyone write one sentence comparing your prediction to the calculated value. Start with: I predicted…'		feedback before Lesson 6.
4 — DRIVING QUESTION BOARD (DQB) UPDATE (10 minutes) 📹 **VIDEO:** Video: Plinko Probability	Students return to the class Driving Question Board. Each home group receives two sticky notes of different colours: a GREEN note for 'What we can now EXPLAIN or CALCULATE about the mkokoteni' and an ORANGE note for 'New questions	Distribute sticky notes. Say: 'Green note — what can you now explain or calculate that you could not at the start of today's lesson? Be specific: use numbers if you can. Orange note — what NEW question has today's lesson given you?' Allow 4 minutes of group	Two-Colour DQB Protocol — Separating 'what we can now explain' (green) from 'new questions raised' (orange) prevents premature closure and models authentic scientific inquiry, where answering one question always generates new ones. The	Observable indicator: Every group's green note contains at least one specific, quantitative claim linked to the mkokoteni or free-fall calculations (e.g., a speed value, a time value, or a named equation). Groups whose green notes are vague ('we learned

Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	this lesson raised'. Groups spend 4 minutes writing their notes. Green note examples expected: 'We can calculate the speed of the mkokoteni at any point if we know the time or distance.' 'After 5 seconds the cart is already doing 180 km/h — faster than a matatu on the highway.' 'A mango falling from 5 m hits the ground at 10 m/s.' Orange note examples expected: 'Does the real cart actually reach these speeds or does friction slow it down?' 'Why did we use a = 10 m/s² — is it exactly 10 everywhere in Kenya?' 'What would a speed–time graph of the mkokoteni look like?' Groups place notes on the DQB. Teacher reads 2–3 aloud and explicitly connects each orange note to the upcoming lessons: 'This question about the graph is exactly what we'll investigate in Lesson 6. Hold onto that question.' Teacher also points to the KICD Key Inquiry Questions posted permanently on the DQB: 'How is free fall motion important in real life?' — and asks: 'Has today's lesson helped us answer this? Who can give one sentence?' Cold-call 2 students.	writing. WAIT — circulate without intervening. After groups post notes, stand at the DQB and read selected notes aloud. For each green note, affirm the connection: 'This group says they can calculate speed from distance — which equation did they use? Shout out.' For each orange note, map it to the lesson sequence: 'Friction — excellent — Lesson 7 will address that. Graph — Lesson 6. Exact value of g at different altitudes — that's a great senior school extension question.' Finally, point to the KICD Key Inquiry Questions and cold-call: 'Chebet, in one sentence, how is free fall motion important in real life?' (10-second wait.) 'Baraka, how can the dangers of free fall on a road like the Limuru escarpment be mitigated?' (10-second wait.) Record both answers verbatim on the board under 'Growing Answers to Our Key Questions'.	public display of growing answers to the KICD Key Inquiry Questions makes cumulative learning visible to the whole class.	about gravity') are prompted: 'Can you add a number to that note — what speed did you calculate?' Teacher photographs the DQB at the end of this phase for the lesson portfolio.
5 — MODEL BUILDING (10 minutes) 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza	Students open their cumulative 'Escarpment Motion Model' — a diagram they have been building since Lesson 1. In earlier lessons they labelled: the mkokoteni at the top (u = 0), arrows showing increasing speed down the slope, and question marks at Points B and C. Today they replace the question marks with calculated values: v = 50 m/s (180 km/h) at t = 5 s / s = 125 m, and v ≈ 100 m/s (360 km/h) at s = 500 m. They also add a new panel to the right of the escarpment diagram showing a	Say: 'Open your Escarpment Motion Model. You have been building this since Lesson 1. Today it gets numbers.' Display the model template on the projector showing the blank question marks at Points B and C. Say: 'Replace every question mark with the value you calculated today. Then add the Kisumu mango inset panel. Finally, write your two-to-three sentence Model Caption explaining WHY the cart keeps accelerating and HOW we predict its speed.' Allow 7 minutes.	Cumulative Annotated Model with Caption Writing — The physical diagram grows lesson by lesson, making conceptual progress visible and spatial. Adding numerical values to the model reinforces the idea that equations are descriptive tools, not abstract procedures. The caption-writing task demands integration of causal explanation (WHY gravity accelerates the cart) with predictive calculation (HOW FAST), directly mirroring the two-part driving question.	Observable indicator: The updated model contains correct numerical values at Points B and C (with units) and the Kisumu mango inset. The Model Caption explicitly names gravity as the cause of acceleration AND references at least one equation by name or symbol. Teacher reviews 5–6 models at the end of the lesson; models that are numerically correct but causally silent ('the cart goes 180 km/h') receive a written prompt: 'What force causes this acceleration? Add one sentence

(PLIX) Source: CK-12 🔍 Search ARES for similar readings	vertical free-fall inset: the Kisumu mango tree with h = 5 m labelled, t = 1 s marked on a dashed fall-path, and v = 10 m/s at impact. Below the diagram, students write a 'Model Caption' of 2–3 sentences: 'Gravity ($g \approx 10$ m/s²) continuously accelerates the mkokoteni because no engine is needed — only an unbalanced gravitational force acting down the slope. Using $v = at$, $s = \frac{1}{2}at^2$ and $v^2 = 2as$ with $u = 0$, we can predict the speed and distance at any moment. After 5 seconds the cart is already travelling at 180 km/h — far faster than any safe road speed in Kenya.' Students who finish early are challenged: 'Add a third scenario to your model: a stone thrown vertically upward from the Rift Valley floor — how would you modify the equations if $u \neq 0$?'	WAIT — circulate and read captions. Prompt students whose captions are procedural only ('we used the equations to get the answers') with: 'Why does the cart accelerate without an engine? Your caption must answer the WHY, not just the HOW.' In the final 3 minutes, cold-call one student to read their Model Caption aloud: 'Nekesa, read your caption to the class.' After reading, ask the class: 'Does this caption answer both parts of our driving question — why it keeps getting faster, AND whether we can predict exactly how fast? Thumbs up if yes.' If thumbs are mixed, invite one student to suggest an improvement. Close the lesson: 'By Lesson 8, your model will also show what happens when friction and braking forces are added. Today you built the mathematical foundation — gravity, free fall, and the three equations of motion. Well done.'		explaining why.' These are returned at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) During the Jigsaw, did Expert C groups ($v^2 = 2as$) struggle more than Expert A and B groups? If so, plan a brief whole-class re-teaching of that equation at the start of Lesson 6 before introducing velocity–time graphs. (2) Did students' Prediction vs. Calculation reflection sentences reveal genuine surprise at the high speeds calculated (180 km/h after only 5 seconds)? If many students simply wrote 'I was wrong', consider adding a structured discussion prompt in Lesson 6: 'What does it FEEL like that the cart is moving at 180 km/h? Where have you seen vehicles moving that fast in Kenya?' to deepen affective engagement with the road safety PCI. (3) Review the 5 randomly collected Checkpoint Cards. If more than 30% of students made unit conversion errors (m/s to km/h), add a 5-minute unit conversion starter to Lesson 6. (4) Were the orange DQB notes dominated by friction questions? This confirms readiness for Lesson 7 (effects of friction and braking). If few students raised friction spontaneously, introduce it explicitly in the Lesson 6 DQB debrief. (5) Did the Model Caption writing reveal that students can explain causality (gravity as the unbalanced force) or only procedure (substituting into equations)? If causal language is absent, plan a targeted 'Force and Motion' concept discussion at the start of Lesson 6 linking Sub-Strand 3.2 back to Newton's Second Law.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students verified $g \approx 10$ m/s² using their own Lesson 4 drop data, confirming that the mkokoteni on the Limuru escarpment and a mango falling from a tree in Kisumu experience the same gravitational acceleration.

What did I learn?	When a body is released from rest ($u = 0$) under gravity ($a = g = 10\ m/s^2$), its motion is fully described by three simplified equations: $v = at$ (speed after time t); $s = \frac{1}{2}at^2$ (distance after time t); $v^2 = 2as$ (speed after distance s). Applying these equations shows that the mkokoteni reaches approximately 50 m/s (180 km/h) after only 5 seconds — far beyond any safe road speed — explaining why witnesses described it as becoming dangerously fast within seconds.
How does this explain the phenomenon?	Gravity is an unbalanced force that continuously accelerates any object released from rest on a slope or in free fall. Because the force never stops acting, speed increases every second by $g \approx 10\ m/s^2$, making the mkokoteni's acceleration inevitable without an engine. The three equations of linear motion transform this physical insight into precise, testable predictions — answering the driving question: we CAN predict exactly how fast and how far the cart will travel before it crashes.

LESSON 6 (80 minutes (2 × 40-minute lessons)): Applications, Road Safety & Real-Life Contexts of Linear Motion	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students apply the equations of linear motion and free-fall principles to investigate braking distances of a matatu at different speeds on Mombasa Road, connect these findings to NTSA road-safety guidelines and steep-road signage (e.g., the Kinangop descent), and research one additional real-life application of linear motion in Kenya — building toward a complete answer to the driving question about the runaway mkokoteni.
Knowledge	Students will know: (1) how to apply the equations of linear motion ($v = u + at$; $s = ut + ½at^2$; $v^2 = u^2 + 2as$) to calculate braking distance under uniform deceleration; (2) that braking distance increases non-linearly (as the square of speed) making high speed disproportionately dangerous; (3) NTSA guidelines and the purpose of warning signs on steep Kenyan roads (Kinangop descent, Limuru escarpment, Timboroa); (4) at least one additional Kenyan real-life context in which linear-motion principles are applied (SGR train scheduling, pole-vault athletics, construction-site safety, etc.).
Skills	Students will be able to: (1) substitute values correctly into $v^2 = u^2 + 2as$ to solve for braking distance; (2) construct and interpret a data table and bar/line graph of speed vs. braking distance; (3) identify variables (initial speed, deceleration) and control them across calculations; (4) present findings as a concise group report with supporting evidence; (5) evaluate road-safety measures using quantitative evidence.
Attitudes	Students will appreciate: (1) that mathematics is a life-saving tool — the same equations used for the mkokoteni apply to every vehicle on Kenyan roads; (2) personal and collective responsibility for road safety; (3) the value of NTSA regulations and steep-road signage as evidence-based interventions; (4) curiosity about linear-motion applications beyond the classroom.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	Lessons 1–5 established that the mkokoteni accelerates because gravity acts continuously on it, and students can now calculate its speed and displacement at any instant. Lesson 6 transfers that mathematical power to a new, urgent context — a matatu on Mombasa Road — revealing that the same physics that makes the cart dangerous also governs every braking event on Kenyan roads. By quantifying braking distance and linking it to NTSA guidelines and road signage, students gain a complete, actionable answer to the driving question and are positioned to present their understanding in Lesson 7.
Safety Notes	No hazardous materials or physical drops in this lesson. If students use personal digital devices, remind them to follow school ICT-use policy and not to access unsafe websites. If printed newspaper articles include graphic crash images, preview and redact before distribution. Students working in groups should share devices respectfully and return all borrowed print resources.

B. LESSON OVERVIEW

In this culminating application lesson, students pivot from the Limuru mkokoteni to a matatu travelling at three different speeds (30 km/h, 60 km/h, and 90 km/h) on Mombasa Road. Using $v^2 = u^2 + 2as$ with a standard deceleration value, groups calculate braking distances, tabulate results, and graph speed vs. braking distance — discovering the non-linear (quadratic) relationship that makes speeding so deadly. The class then analyses NTSA data cards and photographs of warning signs on steep Kenyan roads (Kinangop, Limuru, Timboroa escarpments) to connect the physics directly to road-safety policy. In the second half of the lesson, groups use digital devices or print resources to research one additional Kenyan real-life linear-motion application and prepare a 2-minute group presentation. The lesson closes with a DQB update and a cumulative model revision that now encompasses both free fall and braking — unifying the mkokoteni story with everyday Kenyan road safety.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students are shown a projected image of a matatu (14-seater) on Mombasa Road with a speed-limit sign reading 80 km/h and a second image of the same road with skid marks after a crash. The teacher poses: 'If a matatu is travelling at 30 km/h and the driver brakes hard, it stops in a certain distance. If the matatu is travelling at 90 km/h — three times as fast — predict how many times longer the braking distance will be.' Students write individual predictions in their exercise books using sentence starters: 'I predict the braking distance at 90 km/h will be ___ times longer because ___.' After 3 minutes of silent writing, the teacher cold-calls 4 students (mix of confidence levels) to share predictions without revealing who is 'right'. Teacher records the range of predictions (×3, ×6, ×9) on the board in a 'Prediction Bank' column, telling the class: 'We will return to these predictions once we have calculated the actual values — just like engineers at NTSA do before setting speed limits.'	1. Project the two images and say: 'Look at these two images for 30 seconds. What do you notice? What do you wonder?' [WAIT 30 s — silent observation]. 2. Say: 'Turn to your desk partner. Share one thing you noticed and one question you have. You have 60 seconds.' [WAIT 60 s — pair talk]. 3. Bring the class together and cold-call 3 pairs (6 students) to share, recording keywords on the board (speed, distance, braking, crash). 4. Frame the prediction task: 'Now I want each of you to make a mathematical prediction. Use what you know from our mkokoteni work — specifically the equation $v^2 = u^2 + 2as$ — to guide your thinking. Write it down. 3 minutes, silent.' [WAIT 3 min]. 5. Cold-call 4 students; for each say 'Thank you — tell us your reasoning.' Record all predictions without evaluating. 6. Say: 'Notice that our predictions range from ×3 to ×9. That is a huge difference. The mathematics will tell us who is right — and why it matters for every person in this room who will one day ride a matatu.'	Bridging Analogies — students consciously transfer the mkokoteni equation ($v^2 = u^2 + 2as$) to a new vehicle context before formal calculation. This activates prior knowledge and creates cognitive tension (predictions disagree) that motivates the observation phase.	Teacher scans written predictions for correct identification of the relevant equation and for the direction of reasoning (faster → longer stop). Observable indicator: student writes $v^2 = u^2 + 2as$ or mentions 'square of speed' in their prediction rationale. Students who do not reference an equation are flagged for targeted support during Phase 2 group work.
Phase 2 — Observe 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups of 4 receive a structured data-collection worksheet. The worksheet provides: initial speeds u = 30 km/h, 60 km/h, 90 km/h (converted to m/s on the sheet: 8.33 m/s, 16.67 m/s, 25.00 m/s); final velocity v = 0 m/s (vehicle stops); assumed uniform deceleration $a = -8$ m/s² (standard dry-tarmac braking value, noted as an NTSA engineering estimate). Students use $v^2 = u^2 + 2as$ → s =	1. Distribute worksheets and say: 'You have 18 minutes to complete the calculations, fill the table, and draw the graph. Assign roles now: Calculator, Recorder, Graph-Plotter, Evidence-Analyst. Rotate if you finish early.' 2. Circulate every 3 minutes; at each group pause 10 seconds [WAIT 10 s] before speaking — observe what students are doing before intervening. 3. Prompt groups that	Data-Table and Graph Construction (Science and Engineering Practice: Analyzing and Interpreting Data) — students generate primary quantitative data, visualise it graphically, and observe an emergent pattern (quadratic relationship) that cannot be seen in a single calculation. The NTSA evidence card then connects the mathematical pattern to a real policy artefact, making the	Teacher checks each group's table for correct unit conversion (÷3.6) and correct substitution into $s = -u^2/2a$. Observable indicator: braking distances should be approximately 4.3 m, 17.4 m, and 39.1 m for 30, 60, and 90 km/h respectively. Groups with values significantly different are asked to show their working step-by-step. Graph shape (upward curve, not straight line) is a second

📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	($v^2 - u^2$) / 2a to calculate braking distance for each speed. They record results in a table (columns: speed in km/h, speed in m/s, braking distance in m) and then plot a graph of braking distance (y-axis) vs. speed in m/s (x-axis) on graph paper provided. Groups observe the shape of the curve and annotate: 'The graph is not a straight line — it curves upward.' A second 'evidence card' shows a laminated photograph of a NTSA road sign on the Kinangop descent reading 'LONG STEEP DESCENT — TRUCKS USE LOW GEAR' alongside an infographic showing that 60% of fatal crashes on Kenyan roads involve excessive speed on slopes. Groups discuss: 'How does your graph explain why this sign exists?'	are stuck on unit conversion: 'What does the worksheet say about converting km/h to m/s? Divide by 3.6 — try that and show me.' 4. After 12 minutes, say: 'Pause your calculations. Look at your table so far. What pattern do you see in the braking-distance column as speed doubles?' [WAIT 10 s] Cold-call 2 groups. 5. After 18 minutes, distribute the NTSA evidence card. Say: 'Read the card silently for 2 minutes, then discuss in your group: how does your graph explain this sign?' [WAIT 2 min silent, then 3 min group]. 6. Cold-call 2 groups to share their connection. Record key phrases on the board: 'speed doubles → distance quadruples'; 'non-linear relationship'; 'same physics as the mkokoteni.'	physics consequential.	observable indicator of correct understanding.
Phase 3 — Explain 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class sense-making discussion. The correct braking distances are revealed on a projected table. Students compare with their Phase 1 predictions — the class discovers that at ×3 speed, the braking distance is ×9 longer (confirming the quadratic relationship: $s \propto u^2$). Students articulate why: 'Because in $v^2 = u^2 + 2as$, with $v = 0$, we get $s = u^2/2a$ — braking distance is proportional to the square of initial speed.' The teacher then explicitly connects this back to the mkokoteni: 'At Limuru, the cart starts from rest and accelerates — every second it travels faster, and if it needs to stop, the braking distance would grow as the square of its speed. No one was pushing it, yet gravity kept doing work, building up speed. Our equations now let us calculate exactly how dangerous	1. Project the correct answers table. Say: 'Compare your group's values with these. If yours differ, find the step where the error occurred — you have 2 minutes.' [WAIT 2 min]. 2. Ask: 'By how many times did the braking distance increase when speed tripled from 30 to 90 km/h?' [WAIT 10 s] Cold-call 3 students. Target at least one student who predicted ×3 in Phase 1 — guide them to see: 'You predicted ×3 — the actual answer is ×9. Why the difference?' 3. Write on board: $s = u^2/2a$. Say: 'Read this equation. What does it tell us about the relationship between s and u?' [WAIT 10 s]. Cold-call 2 students. 4. Say: 'Now connect this to the mkokoteni. Use Think-Pair-Share: how does what we found about braking distance connect to the cart on the Limuru escarpment? 2 minutes with your partner.' [WAIT	Argument from Evidence (Science and Engineering Practice: Engaging in Argument from Evidence) — students construct a written argument (Safety Memo) using quantitative data as warrants. This deepens causal understanding (speed² drives danger) and models how scientists communicate findings to policymakers — a real-world scientific practice.	Teacher reads over shoulders during memo writing. Observable indicators: (1) memo includes a specific calculated braking distance; (2) student uses 'proportional to the square' or equivalent phrasing; (3) recommendation is logically linked to the data. Students who write only qualitative statements ('high speed is dangerous') are prompted: 'Which number from your table supports that? Add it in.'

	that was.' Students individually write a 'Safety Memo' (3–4 sentences) addressed to the Nairobi County Roads Department, using at least one calculated value, recommending a maximum speed for the Limuru escarpment descent. Memos are shared in pairs, then 3 are read aloud to the class.	2 min]. Cold-call 2 pairs. 5. Introduce the Safety Memo task. Say: 'You are now a scientist advising the Nairobi County Roads Department. Write a 3–4 sentence memo recommending a speed limit for the Limuru descent. You must include at least one number from today's calculations. 5 minutes, individual, silent.' [WAIT 5 min]. 6. Pairs share memos; cold-call 3 students to read aloud. Highlight strong use of evidence: 'Notice how Wanjiku used the calculated value of 39.1 m to argue her point — that is evidence-based communication.'		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher directs students back to the class Driving Question Board displayed on the wall: 'Why does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?' Each group receives two sticky notes. On the first (green): they write one thing today's lesson has confirmed or answered about the driving question. On the second (orange): they write one new question or connection today's lesson opened up (e.g., 'Does the mass of the mkokoteni affect how far it takes to stop?'; 'What deceleration do cart brakes actually provide?'; 'Does the SGR train use the same equations?'). Groups post notes on the DQB. The teacher reads selected notes aloud, clusters related questions, and narrates how the DQB has grown from Lesson 1's basic 'Why does it speed up?' to today's sophisticated 'How does stopping distance grow with speed, and how does road	1. Point to the DQB: 'Look at where we started in Lesson 1. Read your original questions silently.' [WAIT 20 s]. 2. Say: 'Take your two sticky notes. Green: what has today answered about our driving question? Orange: what new question did today open up? 3 minutes, individual.' [WAIT 3 min]. 3. Groups post notes. Teacher reads 4–5 notes aloud, pausing after each: 'Does anyone else have a similar idea? Raise your hand.' 4. Cluster notes visually on the DQB: 'Here we have a cluster about stopping distance. Here, a cluster about real-life applications. Here, questions about the SGR.' 5. Say: 'Notice how our board now has both answers and new questions. That is exactly how science works — every answer opens new doors. The mkokoteni has taken us from a simple observation to road-safety policy to train scheduling. You have been doing real science.'	Driving Question Board Curation (Science and Engineering Practice: Asking Questions) — revisiting and updating the DQB makes the cumulative arc of the unit visible to students. It externalises metacognition: students see what they now know, what remains open, and how each lesson has moved the class forward.	Teacher reads green sticky notes to check whether students are correctly connecting today's braking-distance findings to the mkokoteni phenomenon. Observable indicator: green note explicitly mentions speed, braking distance, or the equation $v^2 = u^2 + 2as$ in relation to the cart or road safety. Orange notes are used diagnostically to identify misconceptions or genuine extensions for Lesson 7 discussion.

	design account for this?' The teacher points to the orange notes: 'These are your questions — Lesson 7 will give you space to present your full answers and explore these new threads.'			
Phase 5 — Model Building (Group Research Presentations) 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Groups (same as Phases 2–3) are assigned or choose one real-life Kenyan linear-motion application from a provided list: (A) SGR Madaraka Express: train scheduling and safe stopping distances between Nairobi and Mombasa; (B) Athletics: Julius Waweru's pole-vault — modelling the run-up as uniformly accelerated motion; (C) Construction safety on a Nairobi high-rise: calculating the impact speed of a falling object from different floor heights using free-fall equations; (D) NTSA speed cameras on Thika Road: how instantaneous speed is measured and how it links to equations of motion. Using 8 minutes of guided research (digital devices, printed resource cards prepared by the teacher, or textbook extracts), each group prepares a 2-minute verbal presentation answering: (i) What is the real-life context? (ii) Which equation(s) of linear motion apply? (iii) How does this connect to the mkokoteni driving question? Groups present in turn; the class gives structured peer feedback using a 'One Star, One Step' protocol (one strength, one suggestion). The teacher closes by updating a cumulative 'Class Model' diagram on the board — a visual map showing the mkokoteni at the centre, with spokes connecting to: braking distance, free fall, SGR scheduling, construction safety, athletics — all	1. Assign/confirm topics. Say: 'You have exactly 8 minutes to research and prepare. Use your resource card, device, or textbook. Your presentation must answer three questions — they are on your card. I will count down at 5 minutes and 7 minutes.' [WAIT 8 min — circulate, prompt groups that are off-task: 'Which equation will you use? That is your anchor.']. 2. Signal time: 'Presenters, stand. Everyone else: listen for (i) the context, (ii) the equation, (iii) the connection to the mkokoteni.' 3. Each group presents for 2 minutes. After each, say: 'One Star, One Step — group to my left.' [Allow 45 s peer feedback]. 4. After all presentations, draw the cumulative Class Model on the board. Say: 'Point to where each presentation fits on this map.' Cold-call 4 students to physically come and draw a spoke or label a connection. 5. Close: 'Six lessons ago, witnesses at the Limuru escarpment asked why the mkokoteni kept getting faster. Today you have answered that question — and you have shown that the same mathematics explains braking distances, train schedules, falling objects, and athletics. You are now equipped to predict motion. That is the power of physics.' 6. Remind students: 'For Lesson 7, bring your Safety Memo and your group's presentation notes. You will	Cumulative Model Revision (Science and Engineering Practice: Developing and Using Models) — the Class Model diagram is a living artefact updated each lesson. In Lesson 6 it expands from the mkokoteni-centric model to a web of real-life applications, making explicit that one set of equations (the three kinematic equations) explains an entire domain of phenomena. Group presentations add peer-generated evidence to the shared model.	Teacher listens for three indicators during presentations: (1) correct identification of which equation applies (e.g., Group C must identify $v^2 = 2gh$ for free fall from rest); (2) correct substitution of at least one numerical value; (3) explicit verbal connection back to the mkokoteni or driving question. Peer feedback via 'One Star, One Step' is also monitored — students who cannot articulate a strength or suggestion are evidence of surface-level listening and are redirected: 'What equation did that group use? Was it the right one?'

	linked by the same three equations.	synthesise everything into your final answer to the driving question.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) Did students' Safety Memos include quantitative evidence, or did most remain qualitative? If the majority did not cite a specific calculated value, plan a short worked-example opener for Lesson 7 that models evidence-based writing. (2) Were the braking-distance calculations completed accurately by most groups? If unit-conversion errors (km/h → m/s) were widespread, consider providing a unit-conversion reference card in future lessons. (3) Did the group research presentations make a genuine connection to the mkokoteni phenomenon, or did they treat the application as a separate topic? If connections were weak, restructure the presentation template in Lesson 7 to require an explicit one-sentence link. (4) Which orange sticky notes on the DQB represent genuine conceptual extensions (e.g., effect of mass on braking) versus misconceptions (e.g., 'heavier vehicles stop faster')? Plan to address the most common misconception in the first 5 minutes of Lesson 7. (5) Was 8 minutes sufficient for group research, or did groups need more scaffolding on the resource cards? If groups struggled, consider pre-annotating the resource cards with the relevant equation highlighted before the next implementation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students calculated braking distances for a matatu at 30 km/h, 60 km/h, and 90 km/h on Mombasa Road using $v^2 = u^2 + 2as$, graphed speed vs. braking distance (observing a non-linear/quadratic curve), and examined an NTSA evidence card showing a warning sign on the Kinangop descent alongside a statistic that 60% of fatal crashes on Kenyan roads involve excessive speed on slopes.
What did I learn?	Braking distance is proportional to the square of initial speed ($s = u^2/2a$ when $v = 0$), meaning that tripling speed increases stopping distance nine times — the same quadratic relationship that governs how fast the mkokoteni built up dangerous speed on the Limuru escarpment. NTSA road-safety guidelines and steep-road warning signs are direct policy applications of this physics. The three kinematic equations apply equally to the mkokoteni, a braking matatu, an SGR train, a falling construction object, and an athlete's run-up.
How does this explain the phenomenon?	Because braking distance grows as the square of speed (not linearly), small increases in vehicle speed produce disproportionately large increases in stopping distance — exactly as gravity produced a disproportionately dangerous speed increase in the runaway mkokoteni. The same three equations of linear motion ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$) that let us predict the cart's speed at any moment also let engineers calculate safe following distances, set speed limits on escarpment roads, schedule SGR trains, and design construction-site safety protocols — demonstrating that mathematics is a universal, life-saving tool.

LESSON 7 (40 minutes): Group Presentations and Collaborative Model Building	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will synthesise all sub-strand learning by presenting real-life applications of linear motion and collaboratively constructing an annotated explanatory model that connects the mkokoteni phenomenon to motion vocabulary, the four equations, free fall, gravitational acceleration, and road safety mitigation.
Knowledge	Students will demonstrate that the same constant-acceleration equations (v = u + at, s = ut + half at squared, v squared = u squared + 2as, s = half(u+v)t) and the principle of uniform gravitational acceleration (g approximately 10 m per second squared) apply consistently across diverse Kenyan real-life contexts including SGR scheduling, athletics run-ups, construction safety, and braking distances on escarpment roads.
Skills	Students will present research findings clearly and concisely (3-minute group presentations), respond to cold-call follow-up questions, annotate a shared motion diagram or graph sequence, and identify which Driving Question Board questions have been answered by the evidence gathered across Lessons 1 to 7.
Attitudes	Students will demonstrate intellectual courage in responding to unscripted follow-up questions, value collaborative model refinement, and show appreciation for how physics knowledge directly informs life-saving road safety policy in Kenya.
Key Inquiry Question	How is free fall motion important in real life, and how can the dangers of free fall and runaway acceleration on Kenyan roads be mitigated through physics understanding?
Purpose in Storyline	This lesson is the evidence-integration milestone of the sequence. Groups externalise the breadth of linear motion applications gathered in Lesson 6, and the class then fuses all seven lessons of evidence into one coherent annotated model anchored on the mkokoteni phenomenon, making visible the complete chain of reasoning from observation to prediction to safety action. The Driving Question Board is audited so students can see exactly which questions the evidence has answered before the final assessment in Lesson 8.
Safety Notes	

B. LESSON OVERVIEW

The lesson opens with a brief individual prediction task that re-anchors students to the mkokoteni phenomenon and primes them to notice connections during presentations. Groups then present their Lesson 6 research (SGR train scheduling, pole-vault run-up, construction falling-object safety, or similar) in 3-minute slots while classmates use a structured listening frame. The teacher cold-calls two members per group for follow-up questions using precise talk moves and mandated wait times. In the Explain phase the class pools the evidence from all presentations and all previous lessons to collaboratively build and annotate a single visual model — a wall-sized annotated diagram or motion-time graph sequence — showing the mkokoteni descending the Limuru escarpment, labelled with u, v, a, s, t, g, braking distance, and safety mitigation arrows. The DQB is then reviewed: students physically move question cards to an answered column and flag any remaining questions for Lesson 8 consolidation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Graphing Linear Inequalities in the Coordinate Plane: A Sample Application Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students spend 3 minutes writing individual responses in their science journals to a prompt displayed on the board: 'Before you hear the group presentations, predict which ONE application of linear motion that your classmates researched will produce the LARGEST acceleration. Give a numerical estimate using any equation of motion and explain your reasoning in terms of what we know about the mkokoteni on the Limuru escarpment.' Students work silently and commit to a specific application and a numerical value before any presentation begins.	Display the prompt and a 3-minute countdown timer. Circulate silently. Do not answer questions about which groups researched which topics — preserve genuine prediction. After 3 minutes say: 'Keep your journal open. As each group presents, place a tick next to your prediction if evidence supports it, or revise your estimate and note why. This is exactly what a scientist does when new data arrives.' Select 2 students using random name cards to read their predictions aloud before presentations begin so the class has anchors to test.	Individual written prediction creates cognitive commitment and activates prior equation knowledge. Connecting the prediction explicitly to the mkokoteni context ensures the phenomenon remains the explanatory anchor rather than becoming a disconnected exercise.	Teacher scans journals during circulation for quality of numerical estimate and correct equation selection. Students who have not written a numerical value are quietly prompted: 'Which equation connects the variables you know? Write a number, even an estimate — scientists commit before they observe.'
Observe Phase 📹 **VIDEO:** Graphing Linear Inequalities in the Coordinate Plane: A Sample Application Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each group presents its real-life application research in a strict 3-minute slot. Presenters stand at the front; the remaining students use a structured listening frame (three columns in their journals: application name, equation(s) used, connection to the mkokoteni). After each presentation the teacher cold-calls two group members — not necessarily the main presenter — for follow-up questions. At least four groups present; if time allows, a fifth group presents in the final 2 minutes. Classmates complete their listening frame for each group and place ticks or revision notes against their initial prediction.	Time-keep strictly. After each presentation select two members by random name card. Ask: 'You said the SGR train decelerates at 0.5 metres per second squared — which equation did your group use to calculate stopping distance, and what does that tell us about the mkokoteni that had NO brakes at all?' Wait 30 seconds in silence after each question before accepting a response. If the called student is unsure, say: 'Take five more seconds — tell me one variable from the equation that you are confident about and we will build from there.' After all presentations ask the class: 'Which application produced the largest acceleration? Does that match your prediction? Revise your journal entry now.' Use think-pair-share for 90 seconds before taking 3 responses.	Cold-calling two non-presenting members per group distributes accountability and ensures all group members have mastered the content, not just the main speaker. The explicit return question about the mkokoteni after each presentation prevents presentations from becoming isolated from the central phenomenon. The structured listening frame transforms passive observation into active evidence collection.	Teacher listens for correct equation identification and correct use of kinematic variables during cold-call responses. Misconceptions (for example, confusing deceleration with negative velocity rather than negative acceleration) are flagged on the board with the label 'needs model clarification' and addressed in the subsequent Explain and Model Building phases.
Explain Phase 📹 **VIDEO:** Graphing Linear Inequalities in the	The teacher facilitates a whole-class sense-making discussion (approximately 5 minutes) in which students identify the ONE unifying	Ask: 'What is the one thing that ALL the applications we have studied since Lesson 1 have in common — think about the	Requiring students to write their own two-sentence synthesis before seeing the teacher-guided model prevents passive copying.	Teacher reads 4 to 6 journal synthesis statements during the transition to model building. Statements that omit the causal

Coordinate Plane: A Sample Application Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	idea across all presentations. Students are prompted to articulate: 'Every application we heard today — the SGR train, the pole-vaulter, the falling concrete block, the braking matatu — can be described using the same set of five variables and the same four equations as the mkokoteni on the Limuru escarpment. Gravity is the common invisible agent that makes free-fall contexts dangerous and predictable at the same time.' Students write a two-sentence synthesis statement in their journals capturing this unifying idea before the model-building phase begins.	variables, the equations, and the force involved.' Wait 30 seconds. Accept 4 responses using a talk move: 'I heard you say constant acceleration — can someone connect that directly back to our mkokoteni and name the specific force causing that constant acceleration?' Write the agreed unifying statement on the board in student language. Then say: 'Now we are going to build a model that makes this invisible idea visible on paper for anyone who wants to understand why that mkokoteni became dangerous and what could have stopped it.'	The explicit verbal bridge from SGR or athletics back to the mkokoteni re-centres the anchoring phenomenon and reminds students that the DQ has been the organising frame throughout.	role of gravity or that confuse speed and velocity are used (anonymously) as starting points for model revision, with the prompt: 'Our model needs to show not just what happened but WHY — what is missing from this explanation?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** DNA as the "transforming principle" Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The DQB from Lesson 1 is brought to the front. Students individually review the question cards they posted across Lessons 1 to 6. Each student selects at least one card and writes on a sticky dot: 'Answered — evidence from Lesson ___' or 'Still open — needs Lesson 8.' The class physically moves cards into two columns: ANSWERED and STILL INVESTIGATING. Representative students read each answered question aloud and give a one-sentence evidence summary. Questions that remain open — for example, questions about air resistance, the effect of load mass on acceleration, or the exact speed of the mkokoteni before impact — are retained for Lesson 8 consolidation.	Distribute sticky dots. Say: 'You posted these questions in Lesson 1 because they puzzled you. Seven lessons of evidence later — have we answered them? Be honest. A question that is still open is not a failure — it is the engine of the next investigation.' After cards are moved, point to the ANSWERED column and say: 'Look at how much we have built. In Lesson 1 you could not explain why the mkokoteni kept speeding up with no one pushing it. What can you say now?' Wait 30 seconds then cold-call 3 students. Point to the STILL INVESTIGATING column: 'These are the questions Lesson 8 will help us close. Note them in your journals tonight.'	The physical act of moving question cards externalises metacognitive progress — students see, literally, what they now know versus what remains unknown. Requiring an evidence citation (lesson number) prevents superficial card-moving and ensures answers are grounded in specific lesson activities rather than vague recall.	Teacher checks that students can cite specific evidence (an equation, an experiment, a calculation) for each card moved to ANSWERED. Cards moved without evidence citation are placed in a NEEDS EVIDENCE column and revisited at the start of the Model Building phase.
Model Building Phase 📹 **VIDEO:** Graphing Linear Inequalities in the	Working as a whole class on a large sheet of paper pinned to the wall (or on the board), students co-construct an annotated visual model of the mkokoteni descending the Limuru	Assign each group a specific model section before they begin. Say: 'Your group owns the velocity calculation section — write the numbers on the diagram so anyone reading it can predict how	Distributed authorship of the model sections means every student has a tangible contribution to the explanatory product. The velocity calculations embedded directly in the escarpment diagram make the	Teacher photographs completed model sections and checks: (1) v values are numerically correct; (2) v-t graph has correct positive slope labelled as g or a; (3) safety mitigation box references the

Coordinate Plane: A Sample Application Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	escarpment. The model must include: (1) a sketch of the escarpment road with distance markers at 0 m, 50 m, 100 m, 200 m; (2) velocity arrows that grow proportionally at each marker, labelled with calculated v values using v squared = u squared + 2as with a = 10 m per second squared and u = 0; (3) a v-t graph inset showing the linear increase; (4) a FREE FALL inset showing a mango falling from a tree alongside the road, labelled with the same g value; (5) a SAFETY MITIGATION box showing a runaway-truck ramp, warning sign, and the braking-distance equation s = u squared divided by 2a; (6) a REAL-LIFE APPLICATIONS ribbon connecting SGR, athletics, construction, and matatu contexts from the group presentations. Each group contributes one annotated section. The completed model is photographed and printed or redrawn in student journals.	fast the mkokoteni was travelling at 100 m. Make the invisible force visible.' Circulate and ask targeted questions: 'Why does the v arrow get longer as distance increases? Is that linear or quadratic growth? How does your section connect to the free-fall inset?' After 10 minutes of group annotation, facilitate a 3-minute gallery walk around the model. Ask: 'Is there anything our model does not yet explain? Are there any contradictions between sections?' Cold-call 2 students to identify one strength and one gap in the model. Close by saying: 'This model is the answer to our Driving Question — not a final answer, but the best answer our evidence allows right now. Lesson 8 will let you test it one more time individually.'	abstract equations physically situated — students see that v = 44.7 m per second at 100 m means the mkokoteni is travelling faster than a car on a Nairobi highway, which restores the emotional weight of the anchoring phenomenon. The gallery-walk gap identification primes students for Lesson 8 consolidation without pre-teaching the answers.	correct equation; (4) at least one real-life application from the group presentations is connected by an annotated arrow. Groups with errors in velocity calculations are given a corrective prompt card before the lesson ends: 'Recheck your substitution into v squared = u squared + 2as — what value did you use for a and is it consistent with our free-fall experiments in Lesson 4?'

D. TEACHER REFLECTION

After the lesson consider: Did each group presentation advance the central claim that a single set of equations explains diverse Kenyan contexts, or did some groups present applications that remained disconnected from the mkokoteni phenomenon? Were cold-call responses revealing genuine understanding or surface recall — could students explain WHY the equation applies, not just WHICH equation to use? Did the collaborative model genuinely integrate free fall, the equations, and road safety, or did it become a collage of separate topics? Which DQB questions moved to ANSWERED on solid evidence, and which were moved prematurely? Adjust the Lesson 8 problem set to specifically address any equations or concepts that the model revealed as weakly understood. Note any student who struggled to respond under cold-call conditions and build in a paired-response support structure for Lesson 8 if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Before this lesson students could apply individual equations to structured problems but had not yet connected those equations into a unified explanatory account of the anchoring phenomenon. Through group presentations and collaborative model building, students moved from knowing disconnected facts to constructing a coherent causal model: gravity produces constant acceleration, constant acceleration produces quadratic growth in speed and distance, and this quadratic relationship is what makes runaway vehicles on Kenyan escarpments so dangerous so quickly — and also what makes physics an essential tool for predicting and

	preventing such disasters.
What did I learn?	The same four equations of linear motion and the same gravitational acceleration constant (g approximately 10 m per second squared) apply equally to a runaway mkokoteni, a braking SGR train, a falling construction block, and a sprinting athlete. The DQB audit showed that the class can now explain WHY the mkokoteni kept accelerating (gravity provides a constant net force down the slope) and can PREDICT its speed at any point using v squared = u squared + 2as. Road safety measures such as runaway-truck ramps and steep-road warning signs are direct engineering and policy applications of the braking-distance equation.
How does this explain the phenomenon?	

LESSON 8 (80 minutes): Consolidation, DQB Completion & Final Explanation — The Mkokoteni Mystery Solved	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all learning on linear motion by solving structured problems rooted in the anchoring phenomenon, completing the Driving Question Board, and constructing a final scientific explanation for why the mkokoteni accelerates dangerously down the Limuru escarpment.
Knowledge	Students demonstrate mastery of: all four equations of linear motion ($v = u + at$; $s = ut + ½at^2$; $v^2 = u^2 + 2as$; $s = ½(u+v)t$); free fall acceleration ($g = 10\ m/s^2$); and the role of gravitational force as the cause of constant acceleration on a slope.
Skills	Applying equations of linear motion to multi-step quantitative problems; constructing and refining a scientific model; synthesising evidence from the full lesson sequence into a coherent written explanation; evaluating the accuracy of predictions made in Lesson 1.
Attitudes	Appreciating how mathematical models empower communities to predict and mitigate dangerous real-life events such as runaway vehicles on Kenyan escarpment roads; valuing the iterative nature of scientific explanation.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	This is the final lesson of the mkokoteni storyline. Students have built every conceptual tool needed — motion terms, all four equations, free fall, graphical analysis, and the gravitational force model. Today they deploy all of those tools simultaneously to produce a complete, quantitative, evidence-based explanation of the anchoring phenomenon, close every open question on the DQB, and compare their Lesson-1 naive model with their final scientific model. The mkokoteni mystery is fully solved.
Safety Notes	No physical experiment this lesson. Students handle pencils, calculators, and paper only. Ensure adequate desk space for individual assessment work. Remind learners that when discussing road safety on escarpments, language should be sensitive — real accidents on the Limuru–Nairobi road have caused fatalities in Kenyan communities.

B. LESSON OVERVIEW
Lesson 8 is the culminating session of the Sub-Strand 3.2 unit. It opens with a brief re-screening of the original mkokoteni video clip to re-anchor all learning. Students then work individually on a structured 'Final Explanation Assessment' — a set of six problems that revisit every equation and free fall, all framed within the mkokoteni, a Kenyan sprinter, and a falling mango from a tree in Kisumu. After individual work, the class conducts a whole-group DQB review: every sticky-note question posted across Lessons 1–7 is read aloud, answered collectively using the scientific vocabulary and equations built across the unit, and physically moved to the 'We Now Understand' column. Students then complete a journal reflection comparing their Lesson-1 mental model with their final model. The lesson closes with students delivering a 60-second 'Pitch to the Limuru County Engineer' — a verbal final explanation of why the mkokoteni accelerated and what mathematics could have predicted the crash — cementing both content mastery and real-world application.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict: Re-	Students watch the original mkokoteni video clip (or the	Teacher re-screens or narrates the phenomenon: 'Recall our	Cold Recall Writing — students retrieve consolidated knowledge	Observable indicator: Can students spontaneously use the

Anchoring to the Phenomenon (10 minutes) 📹 **VIDEO:** Completion of Number Patterns Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	teacher's verbal re-narration with the still image on the board) for the final time. They then independently write for 3 minutes in their science journals, answering: 'Without looking at your notes, write down everything you now know about WHY the mkokoteni kept accelerating and HOW FAST it was going.' This activates prior knowledge and surfaces any remaining misconceptions before the assessment phase. Students are not allowed to consult notes — this is a cold recall to reveal genuine retention.	mkokoteni on the Limuru escarpment. A cart loaded with maize sacks, released from rest, no engine, no one pushing — yet it becomes lethally fast.' Post the driving question on the board in bold: 'Why does it keep getting faster, and can mathematics predict exactly how fast and how far?' Say: 'In the next 3 minutes, write everything you now know. No notes. This is YOUR thinking after seven lessons.' Start a visible 3-minute timer. After time, cold-call 3 students using name sticks: ask each to share ONE idea. Write their ideas as bullet points under the heading 'What We Now Know — Before Today's Work.' Do NOT correct yet — affirm: 'Hold those ideas. We will test every one of them today.'	from long-term memory without scaffolding. Neuroscientific retrieval practice strengthens retention and reveals genuine understanding versus surface familiarity. Comparing this recall to their Lesson-1 prediction (which will be revisited in Phase 4) makes conceptual growth visible and emotionally meaningful.	words 'gravity', 'constant acceleration', '$g = 10$ m/s²', and at least one equation symbol (v, u, a, s, t) in their unprompted recall? Teacher circulates and places a tick or question-mark next to student journal entries — ticks for correct use of scientific vocabulary, question-marks for entries that still describe acceleration as 'gaining speed because of the hill' without mentioning gravitational force. These question-marks flag students who need targeted support during Phase 2.
Phase 2 — Observe (Individual Assessment): Final Explanation Problem Set (30 minutes) 📹 **VIDEO:** Completion of Number Patterns Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students work individually and silently on the 'Mkokoteni Final Explanation Assessment' — a structured six-problem set distributed as a printed or written sheet. All problems are embedded in Kenyan contexts: PROBLEM 1 (Terms & Definitions — 4 marks): The mkokoteni starts from rest and after 8 seconds reaches a speed of 24 m/s. (a) Define acceleration. (b) Calculate the cart's acceleration. (c) Is this motion uniform or non-uniform? Justify. PROBLEM 2 (Equation 1: $v = u + at$ — 3 marks): A Kenyan sprinter at the Nyayo Stadium accelerates uniformly from rest and reaches 9 m/s in 3 seconds. Calculate: (a) her acceleration; (b) her speed after 5 seconds if she maintains	Distribute the assessment sheet. Say: 'This is your individual work — it is NOT a test for marks today, but treat it seriously because it shows YOU how far you have come since Lesson 1. You have 25 minutes. Begin.' Set a visible timer. Circulate continuously — do NOT answer content questions. If a student is stuck, point silently to the equation reference strip on their desk (a printed strip with all four equations provided as a scaffold — not a full solution guide). After 20 minutes, give a '5-minute warning': 'Five minutes left — if you have not reached Problem 6, skip to it and write at least one sentence.' At 25 minutes, say: 'Pens down. Keep your sheet — we will use it in Phase 3.' Do not collect the sheets yet; students will self-check in Phase 3.	Structured Problem Sequencing (Scaffolded Complexity) — Problems are ordered to mirror the conceptual sequence of the unit: terms → Equation 1 → Equation 2 → Equation 3 → free fall → synthesis. This architecture makes the cumulative learning journey visible within a single task. The final open-response problem (Problem 6) requires students to translate quantitative results into real-world safety recommendations, directly fulfilling the KICD PCIs on road safety and connecting to both key inquiry questions.	Observable indicators during circulation: (1) Students who set up equations correctly but make arithmetic errors — note their names for targeted praise ('You have the right method') during Phase 3 review. (2) Students who use $v = u + at$ when they need $v^2 = u^2 + 2as$ — these students still conflate equation selection; address explicitly in Phase 3. (3) Students who leave Problem 6 blank or write only a qualitative sentence without numbers — flag for extension prompting in Phase 3. Target indicator for 'meets expectations': student correctly solves at least 4 of the 6 problems including Problem 5 (free fall).

	the same acceleration. PROBLEM 3 (Equation 2: $s = ut + ½at^2$ — 4 marks): The mkokoteni is released from rest ($u = 0$) and accelerates at 3 m/s² down the escarpment. (a) How far has it travelled after 6 seconds? (b) At what time has it travelled 54 m? PROBLEM 4 (Equation 3: $v^2 = u^2 + 2as$ — 4 marks): The cart starts from rest and travels 200 m with constant acceleration of 3 m/s². (a) Calculate its speed at the 200 m mark. (b) A witness says the cart was doing 'at least 40 m/s' when it crashed at 267 m — is the witness correct? Show your working. PROBLEM 5 (Free Fall — 4 marks): A ripe mango falls from a tree in Kisumu from a height of 20 m. (a) Taking $g = 10$ m/s², calculate the mango's speed just before it hits the ground. (b) How long does it take to fall? (c) Describe the shape of the speed–time graph for the falling mango. PROBLEM 6 (Synthesis & Mitigation — 3 marks — open response): 'Using the equations of motion and your knowledge of free fall, write TWO recommendations for the Limuru County Engineer that could prevent a future mkokoteni crash. Your answer must include at least one specific number calculated from the equations.' Students work in silence for 25 minutes; the final 5 minutes are used to review their answers.			
Phase 3 —	The teacher leads a whole-class	For Problem 1(b): Write 'a = (v−u)/t	Answer-First Cold Call with	Observable indicators: (1) DQB

Explain: Whole-Class Solution Review & DQB Completion (20 minutes) 📹 **VIDEO:** Completion of Number Patterns Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	walkthrough of the assessment answers. For each problem, one student volunteer (or cold-call) shares their answer and working on the board. The class evaluates: 'Agree or disagree? Why?' Students annotate and correct their own sheets in a different colour pen. After all six problems are reviewed (approximately 12 minutes), the class turns to the Driving Question Board. The teacher reads every remaining sticky-note question posted across Lessons 1–7. For each question, students raise hands to answer using precise scientific language. The sticky note is physically moved (or ticked off) into the 'We Now Understand' column. The teacher closes by asking: 'Is there any question on our DQB we STILL cannot fully answer?' Students consider this honestly — one or two questions about the exact role of friction or air resistance on the mkokoteni may remain, and the teacher acknowledges these as genuine open scientific questions.	$= (24-0)/8 = 3$ m/s²' on the board and ask: 'Is 3 m/s² a reasonable acceleration for a loaded cart on the Limuru road? Think about free fall — g = 10 m/s². Why would the cart's acceleration be less than g?' Wait 12 seconds. Cold-call 2 students. Guide answer: 'Because only a component of gravity acts along the slope — the full 10 m/s² would only occur in vertical free fall.' For Problem 4(b): after revealing $v = \sqrt{0 + 2\times3\times200} = \sqrt{1200} \approx 34.6$ m/s, ask: 'Was the witness right that the cart was doing 40 m/s at 267 m?' Wait 10 seconds. Cold-call: student calculates $v = \sqrt{2\times3\times267} = \sqrt{1602} \approx 40$ m/s. Say: 'The mathematics CONFIRMS the witness. This is the power of equations — they let us verify eyewitness accounts.' For DQB review: point to each sticky note and say: 'Who posted this question? Can YOU now answer it?' If original author answers correctly, lead a brief class clap. For any DQB question that remains partially open (e.g., 'Does air resistance change the acceleration?'), say: 'That is a genuine scientific question that goes beyond Grade 10 Linear Motion — it is answered in fluid dynamics. Well done for asking it.'	Justification — by asking students to commit to an answer before the teacher reveals it, the class maintains high cognitive engagement during review rather than passively copying. The DQB closure ritual makes the arc of learning emotionally visible: students see their Lesson-1 confusion transformed into Lesson-8 understanding, reinforcing scientific identity and self-efficacy. Connecting Problem 4 back to the witness's statement anchors the mathematics to the real phenomenon and models how scientists use equations to evaluate evidence.	closure — at least 90% of posted questions should be answerable by students without teacher prompting; questions that remain unanswered reveal gaps in instruction to inform future teaching. (2) During self-correction, students who write 'I had the right method but wrong arithmetic' vs. students who write 'I used the wrong equation' represent two qualitatively different needs — note these for differentiated follow-up. (3) Listen for correct spontaneous use of cause-language: 'gravity causes the acceleration' rather than just 'gravity makes it go fast.'
Phase 4 — DQB Model Comparison: Lesson 1 vs Final Model (10 minutes) 📹 **VIDEO:** Completion of Number Patterns Source: CK-12 🔍 Search ARES for similar videos	Students open their science journals to Lesson 1's page, where they drew their initial mental model of the mkokoteni phenomenon (a simple sketch with naive labels). They then draw, directly below it, their FINAL model. The final model must include: (1) a labelled force arrow showing gravity (W = mg) acting on the cart; (2) a vector arrow for acceleration (a, pointing downhill); (3) the value a = 3 m/s²	Say: 'Turn back to Lesson 1 in your journals. Find your very first sketch of the mkokoteni — the one you drew before we had done any science.' Pause 20 seconds for students to locate it. Then: 'Below that sketch, draw your FINAL model. It must show five things.' Write on the board: '① Gravity force arrow labelled W=mg ② Acceleration arrow labelled a ③ Numerical value a = 3 m/s² ④ One	Contrastive Model Revision — placing the naive model and the scientific model side-by-side in the same journal page makes conceptual change physically and visually explicit. Research shows that students who articulate what specifically changed in their thinking (not just what they now know) develop deeper epistemic understanding of science as a model-building enterprise. The	Observable indicators in final models: (1) Does the force arrow point downward (gravity) and the acceleration arrow point along the slope? If both point in the same direction vertically, the student has not yet distinguished component vs. full gravitational acceleration — note for follow-up. (2) Is the equation annotation syntactically correct (e.g., 'v = 0 + 3×8 = 24 m/s') or just a formula without

📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	(from Problem 1); (4) at least one equation annotation showing how to calculate speed or distance at any time; and (5) a brief written caption (2–3 sentences) explaining the physics. Students write a one-paragraph 'Reflection on My Thinking Journey' comparing their two models, using the sentence starters: 'In Lesson 1, I thought… Now I understand… The most important idea that changed my thinking was…'	equation annotation ⑤ 2–3 sentence caption.' Set a 7-minute timer. Circulate and spot-check that students are genuinely annotating with physics, not just drawing a prettier cart. After 6 minutes, say: 'Now write your reflection paragraph using the three sentence starters on the board.' After 1 minute, cold-call 2 students to read their reflection aloud. Respond: 'That is exactly what scientists do — revise their models as evidence grows. You have just done real science.'	structured sentence starters prevent vague reflections and require comparative thinking.	substitution? (3) Reflection paragraphs that only describe the cart ('it goes faster') without naming gravity or acceleration as the mechanism have not achieved the target understanding — these students need one-on-one follow-up.
Phase 5 — Model Building & Final Pitch: 60-Second Limuru County Engineer Pitch (10 minutes) 📹 **VIDEO:** Completion of Number Patterns Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each student prepares a 60-second verbal 'Pitch to the Limuru County Engineer' — a concise, evidence-based explanation of the mkokoteni phenomenon and a safety recommendation backed by a specific calculation. Students have 3 minutes to prepare notes (bullet points only — no reading from full sentences). The teacher then cold-calls 3 students to deliver their pitch to the class. The class listens and uses a simple peer-evaluation checklist: ① Did they name gravity as the cause? ② Did they use at least one equation correctly? ③ Did they give a specific, calculated safety recommendation? After the three pitches, the teacher delivers the 'teacher model pitch' (90 seconds) as an exemplar, explicitly naming each of the four equations and both key inquiry questions. The lesson closes with the teacher returning to the driving question and declaring it answered: 'We can now fully explain why the mkokoteni kept getting faster — and we can predict, to the nearest metre per second, exactly how fast it was going at the moment it	Say: 'Imagine you are presenting to the Limuru County Engineer. You have 60 seconds to explain the physics and give one number-backed safety recommendation. Prepare your bullet points — you have 3 minutes.' Start timer. After 3 minutes, say: 'Three volunteers — or I will choose.' Use name sticks to select 3 students if no volunteers come forward. After each pitch, ask the class: 'Checklist — hands up: did they name gravity? Did they use an equation? Did they give a calculated recommendation?' Tally responses publicly on the board. After all three pitches, deliver the teacher model: 'I would tell the engineer: Gravity gives any object on this slope an acceleration of approximately 3 m/s². Using v = u + at, a cart released from rest reaches 30 m/s — 108 km/h — in just 10 seconds. Using s = ut + ½at², it has already travelled 150 m by then. My recommendation: install a sand-filled crash barrier every 100 m starting from the top, rated to absorb at least 30 m/s impact. The mathematics tells us exactly where and how strong.'	Persuasive Scientific Communication (Pitch Format) — the pitch task requires students to synthesise content knowledge, quantitative reasoning, and real-world application into a coherent argument addressed to a non-scientist stakeholder. This mirrors the KICD core competency of 'communication and collaboration' and fulfils the PCI on road safety. The peer-evaluation checklist externalises the success criteria, turning students into active evaluators of scientific explanation quality rather than passive recipients of a teacher summary.	Observable indicators in pitches: (1) Does the student begin with the cause (gravity / gravitational force) rather than a description (it goes fast)? (2) Is at least one equation cited with correct variable identification, even if not calculated aloud? (3) Is the safety recommendation specific and plausibly linked to a calculated number? Students who meet all three criteria are performing at 'meets expectations' or above. Students who describe the phenomenon accurately but cannot link it to an equation are 'approaching expectations' and need one further consolidation task. Record names against these three indicators for summative portfolio evidence.

	crashed.'	Close by revisiting the driving question on the board: 'We started Lesson 1 asking this question. Today, we answered it — completely, quantitatively, and in a way that could save lives on the Limuru road.'		

D. TEACHER REFLECTION
After Lesson 8, reflect on these questions in your professional journal: 1. DRIVING QUESTION RESOLUTION: When you re-read the driving question aloud at the close of the lesson, did students respond with confidence or silence? Could the majority articulate — without prompting — that gravity is the cause of constant acceleration and that the four equations allow precise prediction? If not, which specific equation or concept remained the weakest link across the class? 2. DQB AUDIT: How many sticky-note questions from the full unit (Lessons 1–7) remained unanswered at the end of the DQB review? Were any of these questions that you, as the teacher, could not fully answer within the scope of Grade 10 Linear Motion (e.g., questions about air resistance, friction on the slope, or the cart's rotational motion)? If so, this is a healthy outcome — record these as 'horizon questions' to revisit in future sub-strands (e.g., 3.1 Turning Effect of Force, or Grade 11 Dynamics). 3. ASSESSMENT QUALITY: Review students' Problem 6 open-response answers. Did students recommend specific, calculated interventions (e.g., 'a barrier at 100 m because the cart reaches 24.5 m/s there') or only vague suggestions ('put a speed bump')? The former indicates deep integration of quantitative reasoning with safety thinking; the latter suggests the equations are still experienced as isolated algorithms rather than as tools for real-world decision-making. Plan a brief 10-minute retrieval task at the start of the next sub-strand if fewer than 60% of students gave a quantitative recommendation. 4. MODEL COMPARISON: In students' journal model comparisons (Phase 4), look specifically for: (a) students who drew a correct Lesson-8 model but whose Lesson-1 model shows sophisticated prior knowledge — these students were under-challenged and need extension; (b) students whose Lesson-8 model still lacks a gravity force arrow or shows acceleration pointing upward — these students need a one-on-one remediation conversation before Sub-Strand 3.3. 5. PITCH QUALITY: Were the three student pitches to the 'Limuru County Engineer' substantively different from each other, or did all three use the same equation and the same recommendation? If similar, this suggests students converged on a single 'safe answer' rather than genuinely exploring the problem space. In future, assign different contexts to different students (e.g., one pitches on the sprinter, one on the mango, one on the mkokoteni) to force genuine differentiation. 6. KENYA CONTEXT INTEGRATION: Did the Kenyan contexts (Limuru escarpment, Nyayo Stadium sprinter, Kisumu mango tree) make the problems feel authentic to students, or did they treat them as 'just names'? If the latter, consider beginning future storylines with a community visit or a guest speaker — for example, a Limuru matatu driver or a Kenya National Highways Authority engineer — to ground the physics in lived experience from the very first lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students re-watched the mkokoteni video clip and individually completed a six-problem structured assessment covering all four equations of linear motion and free fall, set within the contexts of the Limuru escarpment cart, a Kenyan sprinter at Nyayo Stadium,

	and a falling mango in Kisumu.
What did I learn?	Gravity provides a constant unbalanced force on the mkokoteni, producing constant acceleration ($a \approx 3$ m/s²). This means speed increases linearly with time ($v = u + at$) and distance increases as the square of time ($s = ut + \frac{1}{2}at^2$). Using $v^2 = u^2 + 2as$, students confirmed that the cart reached approximately 40 m/s (144 km/h) at the crash point — consistent with witness accounts. Free fall ($g = 10$ m/s²) is the limiting case of this same gravitational acceleration when no slope component or friction is present. All four equations are mathematically equivalent tools for predicting motion under constant acceleration.
How does this explain the phenomenon?	The mkokoteni kept accelerating because gravity — an invisible but constant force — continuously transferred energy to the cart throughout its descent, producing a uniform acceleration of approximately 3 m/s². The equations of linear motion are not arbitrary formulas but precise mathematical descriptions of this physical reality, enabling anyone — engineer, driver, or student — to predict the cart's exact speed and position at every moment of its descent, and therefore to design evidence-based safety interventions such as crash barriers placed at calculated distances along the Limuru–Nairobi escarpment road.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.